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Supplement to the NECB 2020: Intent Statements

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Introduction

Background

The national code development system underwent a governance change in November 2022 to support efforts to harmonize construction codes in jurisdictions throughout Canada. The CCBFC, which had been in place since 1991, was dissolved and replaced by a new governance structure in which the Canadian Board for Harmonized Construction Codes (CBHCC) is responsible for developing, approving and maintaining the National Model Codes based on the strategic priorities set by the Canadian Table for Harmonized Construction Codes Policy. The 2020 National Model Codes and related intent statements were developed by the CCBFC.

Understanding the Content of the Supplement to the NECB 2020: Intent Statements

The 2020 editions of the National Model Codes are presented in an "objective-based code" format that is structured in three Divisions (A, B and C). The Codes contain explicitly defined objectives and functional statements (see Division A), which are statements on the functions that the components of a building or facility must perform and the objectives that these functions must satisfy. Most of the Code provisions in Division B—called acceptable solutions—are linked to at least one of those objectives and functional statements. The sets of objectives and functional statements are referred to as "attributions."

Intent statements serve explanatory purposes only and do not form an integral part of the Codes. They contain useful information that helps Code users understand the rationale behind requirements. This contributes to a more accurate interpretation and application of acceptable solutions and a clearer understanding of what alternative solutions should achieve.

Only the provisions in Parts 3 to 8 and 10 of Division B of the National Energy Code of Canada for Buildings (NECB) have intent statements.

For further information on the objective-based code format, including objectives, functional statements and intent statements, users should refer to the preface of the relevant Code.

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Provision: 3.1.1.1.(1)

Intent(s)

Intent 1. To state the scope of Part 3.

Provision: 3.1.1.2.(1)

Intent(s)

Intent 1. To state the application of Part 3.

Provision: 3.1.1.3.(1)

Intent(s)

Intent 1. To state the methods of compliance with Part 3.

Intent 2. To direct Code users to Section 3.2. for the prescriptive path requirements, Section 3.3. for the trade-off path requirements, and Section 3.4. for the building energy performance compliance path requirements.

Provision: 3.1.1.4.(1)

Intent(s)

Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 3.1.1.5.(1)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal characteristics of building envelope materials will be incorrectly determined, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.5.(2)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal characteristics of building envelope materials will be incorrectly determined, which could lead to excessive thermal transfer through the building envelope,

which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.5.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of fenestration and doors will be incorrectly determined, which could lead to underestimation of the overall thermal transmittance values, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.5.(4)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of fenestration and doors will be incorrectly determined, which could lead to underestimation of the overall thermal transmittance values, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.5.(5)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that tests to determine the thermal characteristics of building assemblies other than fenestration and doors will be conducted incorrectly or that the thermal characteristics will be incorrectly determined, which could lead to underestimation of the overall thermal transmittance values, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.6.(1)

Intent(s)
Intent 1. To define which components to include in the calculation of areas of fenestration and doors.

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Provision: 3.1.1.6.(2)

Intent(s)

Intent 1. To define how to measure fenestration made of flat panes not in the same plane or made of curved panes for the purpose of calculating its area.

Provision: 3.1.1.6.(3)

Intent(s)

Intent 1. To define which components to include in the calculation of gross wall area.

Provision: 3.1.1.6.(4)

Intent(s)

Intent 1. To clarify the basis on which to calculate allowable areas of fenestration and doors of additions.

Provision: 3.1.1.6.(5)

Intent(s)

Intent 1. To define which components to include in the calculation of gross roof area.

Provision: 3.1.1.7.(1)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal bridging effect of closely spaced, repetitive structural members and ancillary members will not be accounted for in the calculation of the overall thermal transmittance of assemblies, which could lead to underestimation of the overall thermal transmittance of assemblies, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.7.(2)

Intent(s)

Intent 1. To permit the exclusion of the effects of minor structural elements and building components that completely penetrate the building envelope but do not significantly affect the overall thermal transmittance at the projected area from the calculation of the overall thermal transmittance of assemblies.

Provision: 3.1.1.7.(3)

Intent(s)

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Intent 1. To permit the exclusion of the effects of fasteners from the calculation of the overall thermal transmittance of assemblies, on the basis that fasteners have a limited effect on the thermal transmittance (insulating performance) of assemblies.

Provision: 3.1.1.7.(4)

Intent(s)

Intent 1. To state the overall thermal transmittance of unconditioned enclosures protecting a building envelope component.

Provision: 3.1.1.7.(5)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that all related structural framing will not be included in the calculation of the overall thermal resistance of roof assemblies, which could lead to underestimation of the overall thermal transmittance of the assemblies, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.7.(6)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that roof elements will be considered as wall elements, which could lead to underestimation of their overall thermal transmittance, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that wall elements will be considered as roof elements, which could lead to underestimation of their overall thermal transmittance, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.1.1.7.(7)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the structural framing and perimeter areas of intersecting interior walls will not be considered in the calculation of the overall thermal transmittance of wall assemblies, which could lead to underestimation of the overall thermal transmittance of wall assemblies, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To clarify that the perimeter areas where floor or roof slabs interrupt the wall construction are not to be included in the calculation of the overall thermal transmittance of wall assemblies, on the basis that their effect is addressed in Sentence 3.1.1.7.(8).

Provision: 3.1.1.7.(8)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the structural framing will not be considered in the calculation of the overall thermal transmittance of floor assemblies, which could lead to underestimation of the overall thermal transmittance of floor assemblies, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.1.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the design of the building envelope will increase the overall thermal transmittance of insulation due to air leakage or convective effects, wetting, or moisture bypassing the plane of thermal resistance, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.1.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the design of the building envelope will not adequately account for the effects of air leakage or convection, wetting or moisture bypassing the plane of thermal resistance on the performance of the insulation, which could lead to excessive thermal transfer, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 3.2.1.2.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of the building envelope will be increased where mechanical and electrical system components are placed within and parallel to the building envelope, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.2.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the continuity of insulation will be broken, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.3.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of certain building assemblies will be incorrectly calculated, which could lead to underestimation of the overall thermal transmittance, which could lead to excessive thermal transfer through the environmental separator, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the overall thermal transmittance of certain building assemblies will be unacceptably high, which could lead to excessive thermal transfer through the environmental separator, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.4.(1)

Objective
OE1

Attributions
[F92, F99-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the building will not have a limit on fenestration and door openings through the exterior wall, which could lead to an overall thermal transmittance that is unacceptably high, which could lead to uncontrolled thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.1.4.(2)

Objective
OE1

Attributions
[F92, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the building will not have a limit on skylight openings through the roof, which could lead to an overall thermal transmittance that is unacceptably high, which could lead to uncontrolled thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.2.1.(1)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that an air lock will not be created, which could lead to the infiltration of unconditioned air into conditioned space, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that a door will not return to the closed position after each use, which could lead to the infiltration of unconditioned air into conditioned space, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.2.1.(2)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the configuration of vestibules will require users to open both the interior and exterior doors at the same time, which could lead to uncontrolled transfer of conditioned air through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 3.2.2.1.(3)

Intent(s)

Intent 1. To exempt certain doors from the application of Sentence 3.2.2.1.(1), where

- a vestibule is not justifiable for reasons of function,
- the door is rarely used as a service or emergency exit door,
- the door is only used seasonally,
- the size of the space the door serves is sufficiently small such that the vestibule would occupy an unacceptably large part of the usable space, or
- the building is in a warmer climate zone, and as such, a vestibule would result in minimal energy gains and is therefore not justifiable, and the building is small to mid-size, making the installation of a vestibule uneconomical.

Provision: 3.2.2.2.(1)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of above-ground walls, roofs and floors will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.2.2.(2)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of above-ground walls, roofs and floors in semi-heated buildings will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.2.2.(3)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the overall thermal transmittance will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To clarify that portions of a foundation wall that are above ground and where the top of the foundation wall is less than 0.4 m above the adjoining ground level are permitted to be treated as above-ground walls.

Provision: 3.2.2.2.(4)

Intent(s)

Intent 1. To supersede the requirement of Sentence 3.2.2.2.(1) and require a more stringent overall thermal transmittance where thermal transfer could be increased due to the presence of radiant heating cables, heating or cooling pipes or membranes.

Objective

OE1

Attributions

[F92, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of above-ground walls, roofs and floors with radiant heating cables, heating or cooling pipes or membranes embedded in their surface will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to unnecessary energy consumption for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.2.3.(1)

Intent(s)

Intent 1. To clarify that the fenestration thermal characteristics provided in Article 3.2.2.3. do not apply to doors.

Intent 2. To direct Code users to Article 3.2.2.4. for the thermal characteristics of doors.

Provision: 3.2.2.3.(2)

Objective

OE1

Attributions

[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of fenestration will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To direct Code users to Article 3.1.1.5. for the applicable standards to determine the overall thermal transmittance of fenestration.

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Provision: 3.2.2.3.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of fenestration in semi-heated buildings will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To direct Code users to Article 3.1.1.5. for the appropriate standards to determine the overall thermal transmittance of fenestration.

Provision: 3.2.2.4.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of doors will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To direct Code users to Article 3.1.1.5. for the applicable standards to determine the overall thermal transmittance of doors.

Provision: 3.2.2.4.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of doors in semi-heated buildings will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

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Intent 1. To direct Code users to Article 3.1.1.5. for the applicable standards to determine the overall thermal transmittance of doors.

Provision: 3.2.2.4.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of doors exempted from the requirements of Sentences 3.2.2.4.(1) and 3.2.2.4.(2) will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt certain doors from the requirements of Sentence 3.2.2.4.(1), which would otherwise require all doors to have the same overall thermal transmittance, on the basis that some doors, due to their function, would currently have difficulty meeting the door requirements and some flexibility is needed.

Intent(s)
Intent 1. To direct Code users to Article 3.1.1.6. for the calculation procedures for gross wall area.

Provision: 3.2.2.4.(4)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the nominal thermal transmittance of access hatches will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To clarify that stiffeners and edge construction are not to be included in the determination of the nominal thermal transmittance of access hatches.

Provision: 3.2.2.4.(5)

Intent(s)
Intent 1. To exempt certain types of doors from the requirements of Sentences 3.2.2.4.(1) and 3.2.2.4.(2), on the basis that they typically cannot achieve the performance levels of other doors due to their function and properties.

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Provision: 3.2.3.1.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of below-ground walls or portions thereof will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.1.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of below-ground walls or portions thereof will be unacceptably high, which could lead to excessive thermal transfer through the building envelope of semi-heated buildings, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.1.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulation on walls in contact with the ground will not extend to a sufficient depth, which could lead to the overall thermal transmittance being unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.1.(4)

Intent(s)
Intent 1. To supersede the requirements of Sentences 3.2.3.1.(1) and 3.2.3.1.(2) and require a more stringent overall thermal transmittance where thermal transfer could increase due to the presence of radiant heating cables, heating or cooling pipes or membranes.

Objective
OE1

Attributions
[F92, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.1.(5)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that insulation will not extend horizontally a sufficient distance, which could lead to the overall thermal transmittance being unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.2.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of below-ground roofs will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.2.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of below-ground roofs less than 1.2 m below the exterior ground level will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 3.2.3.3.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of certain floors will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.3.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of certain floors will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.3.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of certain floors-on-ground will be unacceptably high at the perimeter, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.3.(4)

Objective
OE1

Attributions
[F92, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of certain floors-on-ground will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to unnecessary energy consumption for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Intent 2. To limit the probability that the extent of the insulation under certain floors-on-ground will be insufficient in cases where thermal transfer could be increased due to the presence of radiant heating cables, heating or cooling pipes or membranes, which could lead to excessive thermal transfer through the building envelope, which could lead to unnecessary energy consumption for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.3.3.(5)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the overall thermal transmittance of certain floors-on-ground will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt certain installations of insulation from the prescribed requirement to extend the floor insulation vertically around the perimeter of the floor, on the basis that the insulation already provided is sufficient.

Provision: 3.2.4.1.(1)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)
Intent 1. To limit the probability that the building envelope will not be designed and constructed with a continuous air barrier system, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To direct Code users to Article 3.2.4.2. for requirements to achieve compliance using an air barrier system, and to Article 3.2.4.3. for requirements to achieve compliance using air barrier assemblies.

Provision: 3.2.4.2.(1)

Objective
OE1

Attributions
[F90-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the air leakage rate of the air barrier system will be unacceptably high, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the air leakage rate of the air barrier system will not be determined using an appropriate test method, which could lead to an air leakage rate that is unacceptably high, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.2.(2)

Objective

OE1

Attributions

[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate will be incorrectly determined, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(1)

Objective

OE1

Attributions

[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of air barrier assemblies will be unacceptably high, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(2)

Objective

OE1

Attributions

[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that

- the air leakage rate of air barrier assemblies will not be tested using an appropriate test method, or
- air barrier assemblies will not be adequately designed and their air barrier materials will not be tested using an appropriate test method.

This is to limit the probability that the air leakage rate of air barrier assemblies will be unacceptably high, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(3)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of metal and glass curtain walls that act as environmental separators will be unacceptably high and will not be tested using an appropriate test method, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(4)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of fixed windows and skylights that act as environmental separators will be unacceptably high and will not be tested using an appropriate test method, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(5)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of operable windows and skylights that act as environmental separators will be unacceptably high and will not be tested using an appropriate test method, which could lead to uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(6)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of doors that act as environmental separators will be unacceptably high and will not be tested using an appropriate test method, which could lead to

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uncontrolled air leakage through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(7)

Intent(s)

Intent 1. To supersede the requirement of Sentence 3.2.4.3.(6) and allow a higher air leakage rate for revolving doors and automatic commercial sliding doors that act as environmental separators, on the basis that the more stringent rate would compromise their function.

Provision: 3.2.4.3.(8)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To supersede the requirement of Sentence 3.2.4.3.(6) and allow a higher air leakage rate for overhead doors that act as environmental separators, on the basis that the more stringent rate would compromise their function.

Provision: 3.2.4.3.(9)

Intent(s)

Intent 1. To supersede the requirement of Sentence 3.2.4.3.(6) and allow a higher air leakage rate for main entry exterior doors that act as environmental separators, on the basis that their cumulative area is limited and, as such, their impact on overall air leakage is minimal.

Intent 2. To direct Code users to Article 3.1.1.6. for the calculation procedures for gross wall area.

Provision: 3.2.4.3.(10)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)

Intent 1. To limit the probability that the truck box will not be sealed to the building, which could lead to the infiltration of unconditioned air into conditioned spaces, which could lead to uncontrolled air leakage through the environmental separator, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.2.4.3.(11)

Objective
OE1

Attributions
[F90-OE1.1]

Intent(s)
Intent 1. To limit the probability that, when the fireplace is not in use, air movement in the chimney will be unacceptably high, which could lead to uncontrolled air leakage through the environmental separator, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(1)

Intent(s)
Intent 1. To state the scope of the simple trade-off path.

Provision: 3.3.1.1.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the simple trade-off method will be applied to additions or to semi-heated buildings, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(3)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the simple trade-off method will be used to transfer between orthogonal assemblies, which could lead to equating unequal rates of thermal transfer, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(4)

Objective
OE1

Attributions
[F92-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the simple trade-off method will be used to transfer between orthogonal assemblies, which could lead to equating unequal rates of thermal transfer, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(5)

Objective
OE1

Attributions
[F92, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the overall thermal transmittance of building envelope assemblies that have heating or cooling systems embedded in them will be unacceptably high, which could lead to excessive thermal transfer through the building envelope, which could lead to inaccuracy of the trade-off model, which could lead to underestimation of the energy use attributed to the building envelope of the proposed building, which could lead to excessive thermal transfer through the building envelope, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(6)

Objective
OE1

Attributions
[F90, F91, F92-OE1.1]

Intent(s)

Intent 1. To limit the application of the trade-off path.

This is to limit the probability that an increase in performance owing to the use of construction techniques that differ from those prescribed in Section 3.2. will be miscalculated or incorrectly quantified, which could lead to inaccuracy of the trade-off model, which could lead to underestimation of the energy use attributed to the building envelope of the proposed building, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy for heating and cooling, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.1.(7)

Intent(s)

Intent 1. To expand the application of the trade-off path.

This is to permit the trade-off path to be used for building envelope components that are more energy-efficient than those prescribed in Section 3.2., provided the increase in performance can be quantified and is not dependent on occupant interaction.

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Provision: 3.3.1.2.(1)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the sum of the areas of certain assemblies of the reference building will not equal the sum of those same areas of the proposed building, which could lead to inappropriate application of the simple trade-off method, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.3.1.2.(2)

Objective
OE1

Attributions
[F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the energy use attributed to the building envelope of the proposed building will exceed the energy use attributed to the building envelope of the reference building, which could lead to excessive thermal transfer through the building envelope, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 3.4.1.1.(1)

Intent(s)
Intent 1. To direct Code users to Part 8.

Provision: 3.4.1.2.(1)

Objective
OE1

Attributions
[F90, F92-OE1.1]

Intent(s)
Intent 1. To limit the probability that the air leakage rate will be greater than the prescriptive limits for the proposed building using the performance path, which could lead to uncontrolled air leakage through the building envelope, which could lead to inaccuracy of the trade-off model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the design of the building envelope will not adequately account for the performance of the insulation by preventing the increase of overall thermal transmittance due to the effects of air leakage or convection, wetting, or moisture bypassing the plane of thermal resistance

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on the performance of the insulation, which could lead to excessive thermal transfer, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.1.1.1.(1)

Intent(s)

Intent 1. To state the scope of Part 4.

Provision: 4.1.1.2.(1)

Intent(s)

Intent 1. To state the application of Part 4.

Provision: 4.1.1.2.(2)

Intent(s)

Intent 1. To exempt from the application of Part 4

- emergency lighting that is not operated continuously,
- lighting for dwelling units, as automatic controls are not compatible with the functional requirements for dwelling units and much of lighting in dwelling units is provided by plug-in fixtures, which is beyond the mandate of the Code, and
- applications where, due to the nature of the occupancy, it is impractical to apply these requirements.

Provision: 4.1.1.3.(1)

Intent(s)

Intent 1. To state the methods of compliance with Part 4.

Intent 2. To direct Code users to Section 4.2. for the prescriptive path requirements, Section 4.3. for the trade-off path requirements, and Section 4.4. for the building energy performance compliance path requirements.

Provision: 4.1.1.4.(1)

Intent(s)

Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 4.2.1.1.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the power requirements for lighting units for exit signs will be excessive, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.2.(1)

Objective
OE1

Attributions
[F94, F98-OE1.1]

Intent(s)
Intent 1. To limit the probability that ballasts for fluorescent lamps will not meet minimum efficacy levels, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.2.(2)

Objective
OE1

Attributions
[F94, F98-OE1.1]

Intent(s)
Intent 1. To limit the probability that electronic ballasts for fluorescent lamps will not meet minimum efficacy levels, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.3.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that power used for lighting will exceed acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the determination of the interior lighting power allowance will not be based on recognized methods, which could lead to incorrect values of the interior lighting power allowance, which could lead to excessive installed interior lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To direct Code users to Article 4.2.1.4. for the determination of installed interior lighting power, Article 4.2.1.5. for the calculation of interior lighting power allowance using the building area method,

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and Article 4.2.1.6. for the calculation of interior lighting power allowance using the space-by-space method.

Provision: 4.2.1.3.(2)

Intent(s)

Intent 1. In using the space-by-space method, to allow the installed interior lighting power in individual spaces to exceed their interior lighting power allowances, provided that the total of the installed interior lighting power for all spaces in the building does not exceed the total interior lighting power allowance for the building.

Provision: 4.2.1.3.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that incompatible methods of determination of the interior lighting power allowance will be combined, which could lead to incorrect values of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.4.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the determination of installed interior lighting power will not include all relevant luminaires, which could lead to inaccurate determination of installed interior lighting power, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.4.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the determination of installed lighting power will not include all relevant luminaires or will be incorrectly determined for each source, which could lead to inaccurate determination of installed interior lighting power, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.4.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the luminaire wattage will be incorrectly determined, which could lead to inaccurate determination of installed interior lighting power, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.4.(4)

Intent(s)
Intent 1. To exempt from the calculation of installed interior lighting power, those lighting systems whose inclusion in the calculation is considered inappropriate or impractical or would adversely affect the intended functions, or the use of the space or equipment.

Provision: 4.2.1.4.(5)

Intent(s)
Intent 1. To exempt from the calculation of installed interior lighting power, those lighting systems whose inclusion in the calculation would adversely affect the intended functions, or the use of the space or equipment.

Provision: 4.2.1.5.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the building area method will be used to calculate the interior lighting power allowance when it is not appropriate to do so, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.2.1.5.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the primary use of the building will not be used to establish the interior lighting power allowance using the building area method, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.5.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the building area method will be used to calculate the interior lighting power allowance when it is not appropriate to do so, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.5.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the building area method will be used to calculate the interior lighting power allowance by the building area method when there is no appropriate established lighting power density for the building type, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.5.(5)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the determination of the interior lighting power allowance using the building area method will not be properly carried out, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.1.6.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the determination of the interior lighting power allowance using the space-by-space method will not be properly carried out, which could lead to inaccurate determination of the interior lighting power allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the type of control device installed will not be occupancy based, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(2)

Objective
OE1

Attributions
[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the space type used for the selection of interior lighting controls in buildings will not match that used for the determination of LPD by the space by space method, which could lead to an incompatibility in the lighting design, which could lead to unnecessary or excessive operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that interior lighting in buildings will not be adequately controlled, which could lead to unnecessary or excessive operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that lighting controls for interior lighting will be located in an inconspicuous or inaccessible location, which could lead to improper use of the controls, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(5)

Intent(s)

Intent 1. To exempt from the requirement to have manual lighting controls in a readily accessible location those spaces where to do so would compromise safety or security.

Provision: 4.2.2.1.(6)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that lighting in spaces requiring Manual On control will be turned on automatically, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(7)

Intent(s)

Intent 1. To modify the application of Sentence 4.2.2.1.(5) and allow lighting to be turned on automatically in spaces designated for Manual On controls where the exclusive use of Manual On controls would endanger the safety or security of the building occupants.

Provision: 4.2.2.1.(8)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that an excessive amount of the general lighting will be turned on automatically in a space requiring Partial Automatic On control, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that lighting other than general lighting in spaces requiring Partial Automatic On control will be turned on automatically, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(9)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the intermediate setting will be at too high a level for spaces requiring Bi-Level Lighting Control, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(10)

Objective

OE1

Attributions

[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that electric lighting will be used when daylight is available to meet some or all of the illumination needs in enclosed spaces with sidelighting, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(11)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the automatic daylighting controls will not be properly calibrated, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the automatic daylighting controls will not adjust to available daylight in an optimal manner, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 3. To limit the probability that the general lighting in the secondary sidelighted area is not controlled independently of the general lighting in the primary sidelighted area, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(12)

Intent(s)

Intent 1. To exempt from the provisions for automatic daylighting controls of Sentence 4.2.2.1.(10), those locations where sidelighted areas do not provide sufficient daylight to warrant the installation of photocontrols to reduce the energy consumption of general lighting in enclosed spaces, or where a reduction in interior general lighting is not compatible with the function of the space.

Provision: 4.2.2.1.(13)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that electric lighting will be used when daylight is available to meet some or all of the illumination needs in enclosed spaces when skylights or roof monitors are installed, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to

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unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(14)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the automatic daylighting controls will not be properly calibrated, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the automatic daylighting controls will not adjust to available daylight in an optimal manner, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 3. To limit the probability that the general lighting in the overlapping toplighted and sidelighted daylight areas is controlled independently of the general lighting in the daylight area under skylights and roof monitors, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(15)

Intent(s)
Intent 1. To exempt from the provisions for automatic daylighting controls of Sentence 4.2.2.1.(12), those locations where skylights do not provide sufficient daylight to warrant the installation of photocontrols to reduce the energy consumption of general lighting in enclosed spaces.

Provision: 4.2.2.1.(16)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that an excessive amount of the general lighting will be left on in an unoccupied space requiring Automatic Partial Off control, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(17)

Intent(s)

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Intent 1. To exempt from the provisions for Automatic Partial Off controls of Sentence 4.2.2.1.(16), those locations where the energy consumption of the general lighting in enclosed spaces is already low and is reduced to a sufficient level when the space is unoccupied.

Provision: 4.2.2.1.(18)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the general lighting will be left on in an unoccupied space requiring Automatic Full Off control, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(19)

Intent(s)
Intent 1. To exempt from the provisions for Automatic Full Off controls of Sentence 4.2.2.1.(17), those locations where

- shut off is not compatible with the function of the space,
- shut off is not possible as it would endanger the safety or security of individuals, and
- shut off is not possible due to the continuous nature of the operation.

Provision: 4.2.2.1.(20)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that any lighting will be left on in an unoccupied space requiring Scheduled Shut-Off control, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the method of automatically shutting off the lighting will not be appropriate, which could lead to operation of lighting in unoccupied areas, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.2.2.1.(21)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that any one Scheduled Shut-Off control device or system of Sentence 4.2.2.1.(19) will

- control an excessively large area, or
- controls spaces on multiple floor levels.

This is to limit the probability that the overriding of the Scheduled Shut-Off control device will turn lights on in areas that are unoccupied, which could lead to operation of lighting in unoccupied areas, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(22)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that a manual control to override the Scheduled Shut-Off control device or system of Sentence 4.2.2.1.(19) will turn lights on for unnecessary long periods of time or for an excessively large area, which could lead to operation of lighting in unoccupied areas, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.1.(23)

Intent(s)
Intent 1. To exempt from the provisions for Scheduled Shut-Off controls of Sentence 4.2.2.1.(19), those locations where

- shut off is not possible due to the continuous nature of the operation,
- patient care is provided, typically resulting in unpredictability of times of operation, and
- shut off is not possible as it would endanger the safety or security of individuals.

Provision: 4.2.2.2.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the lighting zones in a storage garage will be excessively large, which could lead to frequent detection of activity in the large space, which could lead to infrequent use of the automatic power reduction system, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.2.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the lighting in a storage garage will not be reduced when there is no activity, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.2.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the lighting in covered vehicle entrances and exits from storage garages will not be reduced in transition zones when not needed, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.2.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that lighting power to luminaires serving areas exposed to daylight will not be reduced when there is sufficient daylight, which could lead to the unnecessary operation of lighting systems, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.2.(5)

Intent(s)

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Intent 1. To exempt daylight transition zones and ramps without parking from the provisions for automatic power reduction systems stated in Sentences 4.2.2.2.(1) and 4.2.2.3.(3), on the basis that these areas are typically needed for safety and have minimum impact on the overall power consumption due to lighting.

Provision: 4.2.2.3.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the total primary sidelighted area will be incorrectly determined, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that each primary sidelighted area will be incorrectly determined, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the primary sidelighted width will be incorrectly determined, which could lead to incorrect determination of the primary sidelighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.2.2.3.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the primary sidelighted depth will be incorrectly determined, which could lead to incorrect determination of the primary sidelighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(5)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the total secondary sidelighted area will be incorrectly determined, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(6)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that each secondary sidelighted area will be incorrectly determined, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(7)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the secondary sidelighted width will be incorrectly determined, which could lead to incorrect determination of the secondary sidelighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(8)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the secondary sidelighted depth will be incorrectly determined, which could lead to incorrect determination of the secondary sidelighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.3.(9)

Intent(s)
Intent 1. To exempt from the secondary sidelighted area calculation those areas incapable of receiving adequate daylight.

Provision: 4.2.2.4.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the daylighted area under roof monitors will be incorrectly determined, which could lead to incorrect determination of the toplighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.4.(2)

Objective
OE1

Attributions
[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the determination of the daylighted area under each roof monitor will not appropriately take into account the horizontal projection of daylight onto the area beneath the roof monitor, which could lead to the incorrect determination of the toplighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.5.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylighted area under skylights will be incorrectly determined, which could lead to incorrect determination of the toplighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.5.(2)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the determination of the daylighted area under each skylight will not appropriately take into account the horizontal projection of daylight onto the area beneath the skylight, which could lead to the incorrect determination of the toplighted area, which could lead to not installing appropriate automatic daylighting controls, which could lead to the unnecessary operation of some or all of the general lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.6.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that there will be inadequate controls for lighting, which could lead to improper use of the controls, which could lead to unnecessary operation of lighting, which could lead

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to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.6.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that lighting in guest rooms and suites in commercial temporary lodgings will remain on after the space has been unoccupied for a prolonged period of time, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt bathrooms from the master control requirement as bathrooms may be separately occupied and turning lights off remotely may have safety implications.

Intent 2. To exempt lighting controlled by switched receptacles that are controlled by key captive systems from the requirement as these systems will turn the lighting off immediately upon removal of the key.

Provision: 4.2.2.6.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that there will not be a means of collectively controlling all permanently installed luminaires at the entry to each room in a hotel suite, which could lead to lights being left on when the room is vacated by the occupants, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.2.6.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that there will be inadequate, inconspicuous and inaccessible controls for supplemental task lighting, which could lead to improper use of the controls, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting,

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which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the lighting zone will be incorrectly assigned, which could lead to improperly assigning exterior lighting power allowances, which could lead to exceeding acceptable limits for exterior lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the basic site allowance will exceed acceptable limits, which could lead to excessive use of power for each specific building exterior application, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the connected exterior lighting power for each specific building exterior application will exceed acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for exterior lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(4)

Objective
OE1

Attributions
[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the connected exterior lighting power for all general building exterior applications not listed in Table 4.2.3.1.-C will exceed acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for exterior lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(5)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the connected exterior lighting power for all building exterior applications not listed in Table 4.2.3.1.-D will exceed acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for exterior lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.3.1.(6)

Intent(s)

Intent 1. To exempt certain exterior lighting applications from the maximum exterior lighting power allowances stated in Sentences 4.2.3.1.(1) to 4.2.3.1.(5), on the basis that a restriction on power allowance would negatively impact the ability of the lighting system to perform its function or the designer is incapable of altering the lighting.

Provision: 4.2.4.1.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that exterior lighting will be on when not necessary for the intended use, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.4.1.(2)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

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Intent 1. To limit the probability that exterior lighting designated for dusk-to-dawn operation will be used outside of these hours, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.4.1.(3)

Intent(s)

Intent 1. To exempt certain exterior lighting applications from the requirements for controls stated in Sentences 4.2.4.1.(1) and 4.2.4.1.(2), where such lighting is required for safety, security, or eye adaptation.

Provision: 4.2.4.1.(4)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that exterior lighting will be on at full power when not necessary for the intended use, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.4.1.(5)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that building façade and landscape lighting will be on when not needed based on the designated hours that the nature of the occupancy requires, which could lead to unnecessary operation of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.2.4.1.(6)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that exterior lighting controllers will not be equipped with a means to backup the programming and time setting for an acceptable time period in the event of a power outage, which could lead to loss of programming and time setting, which could lead to unnecessary operation

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of lighting, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.1.1.(1)

Intent(s)

Intent 1. To state the application of the trade-off path for Part 4.

Provision: 4.3.1.2.(1)

Intent(s)

Intent 1. To limit the application of Sentence 4.3.1.1.(1) and not allow the prescriptive requirements of Subsections 4.2.3. and 4.2.4. and Articles 4.2.1.1. and 4.2.1.2. to be traded off.

Provision: 4.3.1.3.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the total annual energy consumption of interior lighting in the proposed building will be incorrectly calculated, which could lead to exceeding acceptable limits for lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.1.3.(2)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the maximum allowed annual energy consumption of interior lighting in the reference building will be incorrectly calculated, which could lead to exceeding acceptable limits for lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.1.3.(3)

Objective

OE1

Attributions

[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that power used for lighting will exceed acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.1.3.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the total annual energy consumption of interior lighting in the proposed building will be incorrectly calculated, which could lead to exceeding acceptable limits for lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.1.3.(5)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the maximum allowed annual energy consumption of interior lighting in the reference building will be incorrectly calculated, which could lead to exceeding acceptable limits for lighting power, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.1.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the installed interior lighting energy will be incorrrectly calculated, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To clarify the meaning of the "installed interior lighting energy."

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Provision: 4.3.2.1.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the annual energy consumption for lighting in a space will be incorrectly calculated, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.2.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the lighting power density for a space will be incorrectly calculated, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.3.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylighted area of a space will be incorrectly determined, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.3.2.3.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylighted area of a space will be determined using more than one daylight source, which could lead to an overestimation of the daylighted area of a space, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.4.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the non-daylighted area of a space will be incorrectly calculated, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.5.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the effective annual daytime operational time of the daylighted area will be incorrectly calculated, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.3.2.5.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the effective annual daytime operational time of the non-daylighted area will be incorrectly calculated, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.5.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the effective annual nighttime operational time will be incorrectly calculated, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.6.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the annual operational time of each space during daytime will be incorrectly determined, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.3.2.6.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the annual operational time of each space during nighttime will be incorrectly determined, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.6.(3)

Intent(s)
Intent 1. To supersede the requirements of Sentences 4.3.2.6.(1) and 4.3.2.6.(2) and allow the annual operational times during daytime and nighttime for each space to be adjusted for the actual days of operation.

Provision: 4.3.2.7.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for daylight harvesting will be incorrectly calculated, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.7.(2)

Intent(s)
Intent 1. To direct Code users to Article 4.3.2.8. for the determination of the daylight supply factor where sidelighting is the primary daylight supply.

Provision: 4.3.2.7.(3)

Intent(s)

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Intent 1. To direct Code users to Article 4.3.2.9. for the determination of the daylight supply factor where toplighting is the primary daylight supply.

Provision: 4.3.2.7.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the daylight system control factor will be incorrectly determined, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.7.(5)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the daylight-dependent control factor for electric lighting will be incorrectly determined, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.8.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the daylight supply factor will be incorrectly calculated, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.3.2.8.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effective luminous transmittance will be incorrectly determined, which could lead to an overestimation of the daylight supply factor, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.8.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the raw daylight supply factor for the rough opening will be incorrectly determined, which could lead to an overestimation of the daylight supply factor, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.8.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor to account for horizontal obstructions will be incorrectly determined, which could lead to an overestimation of the daylight supply factor, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.9.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylight supply factor will be incorrectly determined, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.9.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the mean daylight factor for toplighting will be incorrectly calculated, which could lead to overestimation of the daylight supply factor, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.9.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the room cavity ratio will be incorrectly calculated, which could lead to overestimation of the mean daylight factor for toplighting, which could lead to overestimation of the daylight supply factor, which could lead to an overestimation of the effective annual daytime operational time for the daylighted area, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 4.3.2.10.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for occupancy control will be incorrectly calculated, which could lead to underestimation of the effective annual operational times, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.10.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for personal control will be incorrectly calculated, which could lead to underestimation of the effective annual operational times, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.2.10.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for personal control will be improperly set where the lighting design does not include personal controls, which could lead to underestimation of the effective annual operational times, which could lead to underestimation of the annual energy consumption for lighting in a space, which could lead to underestimation of the installed interior lighting energy, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.1.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the interior lighting energy allowance will be incorrectly calculated, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To clarify the meaning of "interior lighting energy allowance."

Provision: 4.3.3.1.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the annual energy consumption for lighting in a space will be incorrectly calculated, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.2.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the lighting power density for a space will be incorrectly determined, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.3.(1)

Objective
OE1

Attributions
[F94-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the daylighted area of a space will be incorrectly determined, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.4.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the non-daylighted area of a space will be incorrectly calculated, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.5.(1)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the effective annual daytime operational time of the daylighted area will be incorrectly calculated, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.5.(2)

Objective

OE1

Attributions

[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the effective annual daytime operational time of the non-daylighted area will be incorrectly calculated, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead

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to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.5.(3)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effective annual nighttime operational time will be incorrectly calculated, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.6.(1)

Intent(s)
Intent 1. To expand the application of Article 4.3.2.6. to the calculation of annual operational times for the determination of the interior lighting energy allowance.

Provision: 4.3.3.7.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for daylight harvesting will be incorrectly calculated, which could lead to underestimation of the effective annual daytime operational time for the daylighted area, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.7.(2)

Intent(s)
Intent 1. To direct Code users to Article 4.3.2.8. for the determination of the daylight supply factor where sidelighting is the primary daylight supply.

Provision: 4.3.3.7.(3)

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Intent(s)

Intent 1. To direct Code users to Article 4.3.2.9. for the determination of the daylight supply factor where toplighting is the primary daylight supply.

Provision: 4.3.3.7.(4)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylight system control factor will be inappropriately set, which could lead to underestimation of the daylight harvesting factor, which could lead to underestimation of the effective annual daytime operational time for the daylighted area, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.7.(5)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylight-dependent control factor for electric lighting will be inappropriately set, which could lead to underestimation of the daylight harvesting factor, which could lead to underestimation of the effective annual daytime operational time for the daylighted area, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.10.(1)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)

Intent 1. To limit the probability that the factor for occupancy control will be incorrectly calculated, which could lead to underestimation of the effective annual daytime operational time for the daylighted area or overestimation of the effective annual operational times for the non-daylighted and nighttime areas, which could lead to an overestimation of the annual energy consumption for lighting in a space, which

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could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.3.3.10.(2)

Objective
OE1

Attributions
[F94-OE1.1]

Intent(s)
Intent 1. To limit the probability that the factor for personal control will be other than 1, which could lead to underestimation of the effective annual daytime operational time for the daylighted area or overestimation of the effective annual operational times for the non-daylighted and nighttime areas, which could lead to an overestimation of the annual energy consumption for lighting in a space, which could lead to overestimation of the interior lighting energy allowance, which could lead to power used for lighting exceeding acceptable limits, which could lead to excessive use of power, which could lead to unnecessary consumption of energy for lighting, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 4.4.1.1.(1)

Intent(s)
Intent 1. To direct Code users to Part 8 for the building energy performance compliance path for lighting systems.

Provision: 5.1.1.1.(1)

Intent(s)
Intent 1. To state the scope of Part 5.

Provision: 5.1.1.2.(1)

Intent(s)
Intent 1. To state the application of Part 5.

Provision: 5.1.1.2.(2)

Intent(s)
Intent 1. To exempt from the application of Part 5, heating, ventilating or air-conditioning systems or parts thereof where it is impractical to apply these requirements.

Provision: 5.1.1.2.(3)

Intent(s)

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Intent 1. To exempt from the application of Part 5, extensions of heating, ventilating or air-conditioning systems or parts thereof to an addition because of the difficulties and impracticalities of doing so.

Provision: 5.1.1.3.(1)

Intent(s)

Intent 1. To state the methods of compliance with Part 5.

Intent 2. To direct Code users to Section 5.2. for the prescriptive path requirements and Section 5.4. for the building energy performance compliance path requirements.

Provision: 5.1.1.3.(2)

Intent(s)

Intent 1. To limit the method of compliance for back-up systems to the prescriptive path, on the basis that they provide minimum heating, ventilating and air-conditioning levels in the event the primary systems are compromised.

Provision: 5.1.1.4.(1)

Intent(s)

Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 5.2.1.1.(1)

Intent(s)

Intent 1. To direct Code users to the NBC for the minimum sizing requirements for heating, ventilating and air-conditioning systems to ensure that they are not compromised for the sake of energy efficiency.

Provision: 5.2.2.1.(1)

Intent(s)

Intent 1. To direct Code users to the NBC for the minimum design and installation requirements for ducts to ensure that they are not compromised for the sake of energy efficiency.

Provision: 5.2.2.2.(1)

Objective

OE1

Attributions

[F95, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability of unnecessary delivery of conditioned air to areas served by an air distribution system, which could lead to unnecessary consumption of energy for heating and cooling or inefficient use of heating, cooling and ventilating fan systems, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.2.3.(1)

Objective
OE1

Attributions
[F91, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability of uncontrolled air leakage through ducts, plenums and their fittings that form part of a heating, ventilating or air-conditioning system, which could lead to unintended delivery of conditioned air to spaces, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.3.(2)

Objective
OE1

Attributions
[F91, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability of uncontrolled air leakage through ducts, plenums and their fittings that form part of a heating, ventilating or air-conditioning system, which could lead to unintended delivery of conditioned air to spaces, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.3.(3)

Intent(s)

Intent 1. To exempt certain return ducts from the construction, installation and sealing requirements stated in Sentence 5.2.2.3.(1), on the basis that air leakage in such ducts will occur in spaces where there will be no significant energy penalty.

Provision: 5.2.2.3.(4)

Intent(s)

Intent 1. To exempt supply ducts located in the same space to which they supply air from the construction, installation and sealing requirements stated in Sentence 5.2.2.3.(2), on the basis that air leakage from such ducts will occur in spaces where there will be no significant energy penalty and where there is no controlled element whose performance could be negatively affected by the air leakage.

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Provision: 5.2.2.3.(5)

Objective
OE1

Attributions
[F91, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that sealing tape will be used as the primary sealing method, which could lead to failure of the sealant tape, which could lead to unintended delivery of conditioned air to spaces, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.4.(1)

Objective
OE1

Attributions
[F91, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability of uncontrolled air leakage through ducts and their fittings that form part of a heating, ventilating or air-conditioning system, which could lead to unintended delivery of conditioned air to spaces, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.4.(2)

Objective
OE1

Attributions
[F91, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability of excessive air leakage through ducts and their fittings that form part of a heating, ventilating or air-conditioning system as determined by testing, which could lead to unintended delivery of conditioned air to spaces, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.5.(1)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

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Intent 1. To limit the probability that air-handling ducts, plenums and run-outs forming part of a heating, ventilating, or air-conditioning system will be insufficiently insulated, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.5.(2)

Objective
OE1

Attributions
[F93, F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulation on air-handling ducts, plenums and run-outs will be undersized, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.5.(3)

Intent(s)
Intent 1. To exempt certain components of an air-handling system that are in conditioned space from the insulation requirements stated in Sentence 5.2.2.5.(1), on the basis that thermal loss or gain in such components will not have a significant impact on the use of energy.

Provision: 5.2.2.5.(4)

Intent(s)
Intent 1. To exempt certain components of an air-handling system that are in the conditioned space they serve from the insulation requirements stated in Sentence 5.2.2.5.(1), on the basis that thermal loss or gain in such components will not have a significant impact on the use of energy.

Provision: 5.2.2.5.(5)

Intent(s)
Intent 1. To exempt certain components of an air-handling system that are in conditioned space in a dwelling unit and that serve only that dwelling unit from the insulation requirements stated in Sentence 5.2.2.5.(1), on the basis that thermal loss or gain in such components will not have a significant impact on the use of energy and will not affect the energy use of other occupancies.

Provision: 5.2.2.5.(6)

Objective
OE1

Attributions
[F93-OE1.1]

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Intent(s)

Intent 1. To limit the probability that ducts and plenums located outside the building envelope will be insufficiently insulated, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To exempt certain components that are outside of the building envelope and that convey exhaust air that will not have to be reconditioned with building air from the insulation requirements stated in Sentence 5.2.2.5.(1).

Provision: 5.2.2.5.(7)

Intent(s)

Intent 1. To supersede the insulation requirements stated in Sentences 5.2.2.5.(1) and 5.2.2.5.(6) and allow a lower level of insulation in components that have been tested to an accredited standard.

Provision: 5.2.2.5.(8)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that insulation of air-handling ducts, plenums and run-outs will be improperly installed, which could lead to inadequate performance of such insulation in HVAC systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.5.(9)

Objective
OE1

Attributions
[F93, F95, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal performance of the installed insulation will be lower than that specified by the manufacturer, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary demand and/or consumption of energy for heating and cooling, which could lead to inefficient heating and cooling systems, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.2.6.(1)

Objective
OE1

Attributions
[F92, F93, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the insulation on cold-air supply ducts will not have vapour barrier protection, which could lead to reduced performance of the insulation, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.6.(2)

Objective
OE1

Attributions
[F92, F93, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that duct insulation located where it may be damaged or subject to weathering will be unprotected, which could lead to insulation being damaged or weathered, which could lead to reduced performance of the insulation, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.7.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that certain HVAC systems will not use outdoor air to reduce the mechanical cooling energy, which could lead to unnecessary consumption of energy for cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.8.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the mechanical cooling system will not make effective use of outdoor air to supply or augment the supply of cooled air, which could lead to the unnecessary use of mechanical cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.8.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To direct Code users to the ventilation provisions of the NBC for systems that would otherwise reduce the levels of outdoor air to below that prescribed by the NBC for the sake of energy efficiency.

Provision: 5.2.2.8.(3)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that the mix of outdoor air with return air will not be compatible with the temperature required to condition the space, which could lead to the unnecessary use of mechanical cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.8.(4)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that systems with a large cooling capacity will not be designed with lesser stages, which could lead to excessive cycling of the cooling system, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.8.(5)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

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Intent 1. To limit the probability that systems with a significant cooling capacity will not be designed with lesser stages, which could lead to excessive cycling of the cooling system, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.8.(6)

Intent(s)

Intent 1. To exempt certain systems that are susceptible to coil frosting from the requirements of Sentence 5.2.2.8.(3) regarding the mix of outdoor air with return air.

Provision: 5.2.2.9.(1)

Objective

OE1

Attributions

[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the design of the cooling system will not be capable of supplying all of the cooling demand at a reasonable outdoor temperature, which could lead to the unnecessary use of mechanical cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.2.9.(2)

Objective

OE1

Attributions

[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the design of the cooling system will not be capable of supplying all of the cooling demand at a reasonable outdoor temperature, which could lead to the unnecessary use of mechanical cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.1.(1)

Intent(s)

Intent 1. To state the application of Subsection 5.2.3.

Provision: 5.2.3.1.(2)

Objective

OE1

Attributions

[F95, F97-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the power demand of a fan system will be incorrectly calculated, which could lead to underestimation of the power demand, which could lead to inefficient design of the fan system, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.2.(1)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that constant-volume fan systems will be designed with motors that consume excessive power, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.3.(1)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that variable-air-volume fan systems will be designed with motors that consume excessive power, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.3.(2)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that variable-air-volume systems with a significant power demand will not be equipped with controls to sufficiently reduce the power demand when the air delivery volume is reduced, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.3.(3)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)

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Intent 1. To limit the probability that variable-air-volume systems with a large power demand will not be equipped with controls to sufficiently reduce the power demand when the air delivery volume is reduced, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.4.(1)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that enclosed semi-heated and conditioned spaces where fuel-powered vehicles or mobile fuel-powered equipment and appliances are occasionally used will be over-ventilated when not required, which could lead to unnecessary consumption of energy for heating, cooling and electrical equipment and devices, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.3.4.(2)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that commercial foodservice facilities will be over-ventilated when not required, which could lead to unnecessary consumption of energy for heating, cooling and electrical equipment and devices, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.4.1.(1)

Objective
OE1

Attributions
[F91, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability of uncontrolled exhaust of conditioned air to other than conditioned spaces, which could lead to increased demand for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability of uncontrolled intake of outdoor air, which could lead to increased demand for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.4.1.(2)

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Intent(s)

Intent 1. To exempt air intakes and outlets from the damper requirement of Sentence 5.2.4.1.(1) where dampers are not permitted.

Provision: 5.2.4.1.(3)

Intent(s)

Intent 1. To exempt air intakes and outlets from the damper requirement of Sentence 5.2.4.1.(1) where, due to the continuous operation of the HVAC system, they will not be used.

Provision: 5.2.4.1.(4)

Intent(s)

Intent 1. To supersede the requirement of Sentence 5.2.4.1.(1) and allow manual operation of air intake dampers and gravity or spring-operated control of air outlet dampers for small ducts and openings, on the basis that these alternative methods are sufficient to control the air intake or outflow.

Provision: 5.2.4.2.(1)

Objective
OE1

Attributions
[F90, F91, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that air intakes and outlets will remain open to the outside when not in operation, which could lead to unwanted and uncontrolled air leakage through the building envelope, which could lead to increased demand for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.4.2.(2)

Objective
OE1

Attributions
[F90, F91, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability of excessive airflow through closed motorized dampers, which could lead to unwanted and uncontrolled air leakage through the building envelope, which could lead to increased demand for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.4.2.(3)

Objective
OE1

Attributions
[F92, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability of insufficient insulation in ducts located between dampers and the building envelope, which could lead to excessive thermal transfer through the ducts, which could lead to increased demand for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.4.2.(4)

Intent(s)

Intent 1. To exempt certain dampers serving outside equipment from the location requirement of Sentence 5.2.4.2.(1), on the basis that the location of these dampers will not affect thermal transfer or air leakage across the building envelope.

Provision: 5.2.5.1.(1)

Intent(s)

Intent 1. To direct Code users to the NBC for the design and installation of piping for heating, ventilating and air-conditioning systems.

Provision: 5.2.5.2.(1)

Objective
OE1

Attributions
[F95, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that hydronic systems will not be designed so that they can be balanced, which could lead to improper flow distribution, which could lead to inefficient operation of system pumps, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(1)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

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Intent 1. To limit the probability that there will be insufficient insulation on piping in HVAC systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(2)

Intent(s)

Intent 1. To exempt pipes located within conditioned space in a dwelling unit and serving only that dwelling unit from the insulation requirements of Sentence 5.2.5.3.(1), on the basis that thermal losses or gains from such pipes will not have a significant impact on the use of energy and will not affect the energy use of other occupancies.

Provision: 5.2.5.3.(3)

Intent(s)

Intent 1. To exempt from the insulation requirements of Sentence 5.2.5.3.(1) piping located within a conditioned space that conveys fluids with operating temperatures that do not differ significantly from the ambient temperature, on the basis that the thermal loss or gain from the fluid is insignificant.

Provision: 5.2.5.3.(4)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that there will be insufficient insulation for piping in heating and cooling systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(5)

Intent(s)

Intent 1. To supersede the insulation requirement of Sentence 5.2.5.3.(1) and allow a decrease in the thickness of the insulation where its conductivity is lower than that prescribed in Table 5.2.5.3.

Provision: 5.2.5.3.(6)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the thermal conductivity of pipe insulation at the mean temperature rating will not be determined in accordance with an acceptable standard, which could lead to insufficient insulation for piping in heating and cooling systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(7)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that piping insulation will be improperly installed, which could lead to inadequate performance of insulation for piping in heating and cooling systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(8)

Objective
OE1

Attributions
[F93, F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the insulation on piping will be undersized, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.3.(9)

Attributions
[F93, F95, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal performance of the installed insulation will be lower than that specified by the manufacturer, which could lead to uncontrolled thermal transfer losses or gains, which could lead to unnecessary demand and/or consumption of energy for heating and cooling, which could lead to inefficient heating and cooling systems, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.5.4.(1)

Objective
OE1

Attributions
[F92, F93, F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulation on chilled-fluid piping will not have vapour barrier protection, which could lead to reduced performance of the insulation, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.5.4.(2)

Objective
OE1

Attributions
[F92, F93, F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that piping insulation located where it may be damaged or subject to weathering will be unprotected, which could lead to insulation being damaged or weathered, which could lead to reduced performance of the insulation, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.6.1.(1)

Intent(s)
Intent 1. To state the application of Subsection 5.2.6.

Provision: 5.2.6.1.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that the total pump system motor nameplate power will be incorrectly determined, which could lead to underestimation of the power demand, which could lead to inefficient design of the pump system, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.6.2.(1)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that pumping systems that serve control valves will be required to do more work than is needed to meet the load demand, which could lead to unnecessary power consumption, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.6.2.(2)

Intent(s)
Intent 1. To exempt from the requirements of Sentence 5.2.6.2.(1), systems

- that could be damaged if the requirements of Sentence 5.2.6.2.(1) were applied,
- in which the application of Sentence 5.2.6.2.(1) will likely not result in any appreciable energy savings, or
- that control the supply temperature rather than the fluid flow rate.

Provision: 5.2.6.3.(1)

Objective
OE1

Attributions
[F95, F97, F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that oversized hydronic system pumps will be used, which could lead to unnecessary power consumption, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.7.1.(1)

Objective
OE1

Attributions
[F95, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that equipment that is not designed for installation outdoors or in unconditioned spaces will be installed in such conditions, which could lead to

- damage or weathering of the equipment, which could lead to improper operation of the equipment, which could lead to the unnecessary consumption of energy, or
- the use of equipment with improper insulation, which could lead to uncontrolled thermal transfer to or from the system.

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This is to limit the probability of excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.1.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that heating and cooling systems will not include a means to control the temperature, which could lead to overheating or overcooling of the space, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.1.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that each dwelling unit in a building will not include a means to control the temperature in that specific dwelling unit, which could lead to overheating or overcooling of the dwelling unit, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.2.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that each room in a dwelling unit will not include a means to control the heating temperature, which could lead to overheating of the room, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.2.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

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Intent 1. To limit the probability that each room in a dwelling unit will not include a means to control the cooling temperature, which could lead to overcooling of the room, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.3.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that heating and cooling systems will not be equipped with a means to automatically control the temperature when the guest room or suite is not occupied, which could lead to overheating or overcooling of the space, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.4.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that sensors of wall-mounted thermostats will be installed

- either too high or too low,
- on exterior walls that are insufficiently insulated to prevent the sensors from being affected by the outside temperature,
- in direct sunlight or near sources of heat, or
- in drafts or dead air pockets.

Intent(s)
Intent 1. To exempt sensors for thermostats from the prescribed installation requirements in installations where accessibility must be facilitated.

Provision: 5.2.8.5.(1)

Objective
OE1

Attributions
[F95, F97, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that heat pumps with a supplementary heating system will be installed without controls, which could lead to unnecessary operation of the supplementary heaters, which

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could lead to unnecessary consumption of energy for heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the controls will not prevent operation of the supplementary heaters when the load can be met by the heat pump, which could lead to unnecessary operation of the supplementary heaters, which could lead to unnecessary consumption of energy for heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To supersede the operational requirements of the heat pump controls and allow supplementary heaters to operate when the heat pump is in a defrost cycle, on the basis that requiring operation of the heat pump at these times could result in damage to the unit.

Provision: 5.2.8.6.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating or cooling supply to a zone will not be controlled, which could lead to overheating or overcooling the zone, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.6.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that individual perimeter systems designed to offset only building envelope heat losses or gains will be improperly designed and installed, which could lead to overheating or overcooling, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.6.(3)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that the controls will call for simultaneous heating and cooling of a space, which could lead to increased heating or cooling demand, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.8.6.(4)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limt the probability that thermostats for electric resistance heaters will not be capable of performing the control of heating to an acceptable standard, which could lead to overheating of the space, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.6.(5)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that the amount of heating to a vestibule will not be controlled, which could lead to overheating or overcooling of the vestibule space, which could lead to excessive loss of over-conditioned air when doors to the outside are opened, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.7.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that ice- and snow-melting heaters will not be capable of shutting down or being shut down when they are not required, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.8.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that a supply air handler will not be equipped with adequate controls, which could lead to

- heating previously cooled air,

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- cooling previously heated air, or
- excessively heating outdoor air.

This is to limit the probability of unnecessary consumption of energy for heating, ventilation or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.8.(2)

Intent(s)

Intent 1. To supersede the control requirements for supply air handlers stated in Sentence 5.2.8.8.(1) in situations where reheating the supply air is required.

Provision: 5.2.8.8.(3)

Intent(s)

Intent 1. To supersede the control requirements for supply air handlers stated in Sentence 5.2.8.8.(1) in situations where reheating the supply air will not increase the consumption of energy.

Provision: 5.2.8.9.(1)

Objective

OE1

Attributions

[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that a system that controls the temperature of a space by reheating previously cooled air will operate without the proper controls to automatically adjust the temperature of the cool air supply to the highest temperature that will satisfy the temperature-control zone requiring the coolest air, which could lead to the unnecessary use of mechanical cooling to meet the air temperature set-point, which could lead to unnecessary consumption of energy for cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.9.(2)

Objective

OE1

Attributions

[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that a system that controls the temperature of a space by recooling previously heated air will operate without the proper controls to automatically adjust the temperature of the warm air supply to the lowest temperature that will satisfy the temperature-control zone requiring the warmest air, which could lead to the unnecessary use of mechanical heating to meet the air temperature set-point, which could lead to unnecessary consumption of energy for heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.8.9.(3)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that a system that controls the temperature of a space by mixing heated supply air and cooled supply air will not

- automatically adjust the temperature of the warm supply air to the lowest temperature that will satisfy the temperature-control zone requiring the warmest air, and
- automatically adjust the temperature of the cool supply air to the highest temperature that will satisfy the temperature-control zone requiring the coolest air.

This is to limit the probability of unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.8.9.(4)

Intent(s)
Intent 1. To exempt from the requirements for controls stated in Sentences 5.2.8.9.(1), 5.2.8.9.(2) and 5.2.8.9.(3), systems that reduce the air supplied to a minimum in relation to the design flow rate and the ventilation requirements of the temperature-control zone, on the basis that controls will have a negligible impact on the consumption of energy due to the small volume of supply air.

Provision: 5.2.9.1.(1)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that a system that maintains specific humidity by adding or removing moisture will not be equipped with an automatic control device, which could lead to adding or removing moisture beyond the design set point, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.1.(1)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability of exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste energy, which could lead to unnecessary consumption of energy

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for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To exempt from the requirement to have energy recovery capability, systems where the potential for energy recovery from exhaust air is low.

Provision: 5.2.10.1.(2)

Objective

OE1

Attributions

[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heat recovered from energy recovery systems will not be used in building systems, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.1.(3)

Intent(s)

Intent 1. To exempt from the requirement to have energy recovery capability, systems that exhaust air containing fumes, vapours or dust required to be exhausted from the building for reasons of health or safety.

Provision: 5.2.10.1.(4)

Objective

OE1

Attributions

[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that energy recovery systems will not perform at an acceptable level, which could lead to exhausting of conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste energy, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.1.(5)

Objective

OE1

Attributions

[F95, F100-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the effectiveness of energy recovery systems will not be determined using an appropriate test method, which could lead to the exhausting of conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste energy, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.1.(6)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that energy recovery systems will be improperly operated, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.2.(1)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that swimming pools within conditioned spaces will not be equipped with a heat-recovery system capable of recovering a minimum amount of sensible heat from exhaust air, which could lead to unnecessary rejection of reusable waste energy, which could lead to unnecessary consumption of energy for heating or cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To exempt small indoor swimming pools from the requirement for the installation of a heat-recovery system, on the basis that heat recovered from the exhaust air will not be sufficient to have a significant impact on the energy performance of the building.

Provision: 5.2.10.2.(2)

Intent(s)

Intent 1. To exempt from the requirement for the installation of a heat-recovery system, indoor swimming pools where air is dehumidified by a means that does not exhaust air to the outside.

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Provision: 5.2.10.3.(1)

Objective
OE1

Attributions
[F95, F96, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that a refrigeration system will not incorporate sufficient means of recovering the heat it rejects to satisfy some or all of the space-heating or service water heating requirements, which could lead to the unnecessary consumption of energy for heating or service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.4.(1)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that a self-contained mechanical ventilation system serving a single dwelling unit will be operated without heat-recovery capability on its principal exhaust component, which could lead to exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste heat, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt from the requirement to equip the principal exhaust with a heat-recovery system, buildings in climate zones for which the installation of a heat-recovery system is not economically justified.

Provision: 5.2.10.4.(2)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that the performance of heat-recovery ventilators will fall significantly below expectations, which could lead to exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste heat, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.10.4.(3)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that testing the sensible-heat-recovery efficiency of heat-recovery systems will be performed under improper conditions, which could lead to the performance of heat-recovery ventilators falling significantly below expectations, which could lead to exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste heat, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.4.(4)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that the performance of heat-recovery systems other than heat-recovery ventilators will fall significantly below expectations, which could lead to exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste heat, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.10.4.(5)

Objective
OE1

Attributions
[F95, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that the performance of heat-recovery systems in multi-unit residential buildings will be inadequate, which could lead to exhausting conditioned air to the outside, which could lead to the unnecessary rejection of reusable waste heat, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.1.(1)

Objective
OE1

Attributions
[F95-OE1.1]

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Intent(s)

Intent 1. To limit the probably that heating, ventilating or air-conditioning systems will be installed and operated without automatic controls, which could lead to heating, ventilating or air-conditioning systems supplying thermal energy to spaces to maintain occupant comfort when they are unoccupied, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To exempt occupancies other than dwelling units from the requirement for automatic controls where heating, ventilating and air-conditioning systems are intended to operate continuously, on the basis that there will be no periods of non-use of the spaces served.

Intent 2. To exempt from the requirement for automatic controls, zones in buildings other than dwelling units that have a heating or air-conditioning system of less than 5 kW capacity, on the basis that the use of controls will have no significant impact on the energy used.

Provision: 5.2.11.1.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probably that a heating, ventilating or air-conditioning system will be designed and installed without the necessary controls capable of

- shutting down fan systems, heating and cooling equipment and/or auxiliaries, where appropriate, when the space does not require conditioning,
- setting back the space-heating temperature set-point when appropriate for operational needs,
- setting up the space-cooling temperature set-point when cooling system operation is required during periods when the space is not in use,
- reducing or shutting off outdoor air intake during heating or cooling system operation when the space is not in use, which could lead to the unnecessary use of heating or cooling energy to temper outside air, and
- in the case of heat pumps, temporarily suppressing electrical back-up or adaptive anticipation of the recovery point, which could lead to the unit resorting to supplementary heat at the time of recovery.

This is to limit the probability of unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.1.(3)

Intent(s)

Intent 1. To supersede the requirement of Sentence 5.2.11.1.(1) and allow zones with a small heating, ventilating and cooling capacity to have manual controls, on the basis that the use of manual versus automatic controls will have no significant impact on the consumption of energy in these spaces.

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Provision: 5.2.11.1.(4)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)
Intent 1. To limit the probability that controls required by Sentence 5.2.11.1.(1) will be improperly designed such that

- lowering a heating thermostat set-point triggers cooling energy to be used to reach the lowered heating setting, or
- raising a cooling thermostat set-point triggers heating energy to be used to reach the raised cooling setting.

This is to limit the probability of the unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(1)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that air distribution systems will not be divided into individual airflow control areas, which could lead to a lack of airflow control, which could lead to

- an excessive air supply to areas whose instantaneous load can be satisfied with less air,
- excessive air exhausted from those areas, or
- a load demand from areas not served by the zoned airflow.

This is to limit the probability of unnecessary fan energy usage and excessive thermal energy transfer, which could lead to unnecessary consumption or demand of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(2)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that air distribution systems will serve an excessively large area, which could lead to a lack of aircontrol, which could lead to

- excessive air supply to areas whose instantaneous load can be satisfied with less air,
- excessive air exhausted from those areas, or
- load demand from areas not served by the zoned airflow.

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This is to limit the probability of unnecessary fan energy usage and excessive thermal energy transfer, which could lead to unnecessary consumption or demand of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(3)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that airflow control areas required by Sentences 5.2.11.2.(1) and 5.2.11.2.(2) will include temperature-control zones not intended to be operated simultaneously, which could lead to air distribution systems running to satisfy only a portion of the temperature-control zones, which could lead to cycling of more air than required to meet the temperature set-point, which could lead to unnecessary consumption or demand of energy for heating and cooling and fan motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(4)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that airflow control areas required by Sentences 5.2.11.2.(1) and 5.2.11.2.(2) will exceed the control capability of one air distribution system due to a potentially large variance in local conditions, which could lead to cycling of more air than required to meet the temperature set-point, which could lead to unnecessary consumption or demand of energy for heating and cooling and fan motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(5)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)
Intent 1. To limit the probability that airflow control areas will not be equipped with adequate controls, which could lead to heating and cooling equipment performing their function during periods of non-use of the spaces served by the systems, which could lead to unnecessary consumption or demand of energy for heating and cooling and fan motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.11.2.(6)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that an air distribution system will be designed so that a reduction in air delivery does not result in a proportional reduction in fan power, which could lead to unnecessary consumption or demand of energy for heating and cooling and fan motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(7)

Objective
OE1

Attributions
[F95, F97, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the operation of all fan systems and associated primary heating and cooling equipment may become unstable for any length of time while serving a single airflow control area, which could lead to adjustment of controls to non-optimal levels, which could lead to unnecessary consumption or demand of energy for heating and cooling and fan motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.2.(8)

Intent(s)

Intent 1. To exempt temperature-control zones from the requirement to be incorporated into airflow control areas where outdoor air and exhaust make their incorporation impractical.

Provision: 5.2.11.3.(1)

Objective
OE1

Attributions
[F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that pumping stations that are used on a seasonal basis will not be equipped with controls, which could lead to their operation when not warranted, which could lead to unnecessary consumption of energy for pump motors, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.11.4.(1)

Objective
OE1

Attributions
[F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that systems with multiple boilers will not incorporate a means for preventing heat loss when any or all are not operating, which could lead to uncontrolled heat loss through the boilers, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.4.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that boiler plants with a moderate load will not have a means to efficiently address part load, which could lead to unnecessary consumption of energy for heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.4.(3)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that boiler plants with a large load will not have a means to efficiently address part load, which could lead to unnecessary consumption of energy for heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.5.(1)

Objective
OE1

Attributions
[F95, F98-OE1.1]

Intent(s)

Intent 1. To limit the probability that chilled or hot-water systems will be installed without loop temperature reset, which could lead to over-heating or over-cooling of spaces, which could lead to unnecessary consumption of energy for heating and cooling and to inefficient operation of equipment, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.11.5.(2)

Intent(s)

Intent 1. To exempt from the automatic control requirements of Sentence 5.2.11.5.(1), chilled- or hot-water systems that would not operate efficiently using loop temperature reset control.

Provision: 5.2.11.5.(3)

Intent(s)

Intent 1. To exempt from the automatic control requirements of Sentence 5.2.11.5.(1), chilled- or hot-water systems that use variable flow to control capacity.

Provision: 5.2.12.1.(1)

Objective

OE1

Attributions

[F95, F98, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that heating, ventilating and air-conditioning equipment and components will not meet minimum acceptable levels of performance, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.12.2.(1)

Intent(s)

Intent 1. To exempt equipment covered by Article 5.2.12.1. from the requirements in Table 5.2.12.2.

Intent 2. To state the scope of Article 5.2.12.2.

Provision: 5.2.12.2.(2)

Objective

OE1

Attributions

[F95, F97, F98, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that heat rejection equipment will not meet minimum acceptable levels of performance, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for cooling, which could lead to an unacceptable effect on the environment.

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Provision: 5.2.12.2.(3)

Objective
OE1

Attributions
[F95, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that constant speed fans will not be installed with cycling controls, which could lead to unnecessary consumption of energy for cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.12.3.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that systems with components from more than one manufacturer will not be designed in accordance with good engineering practice based on correct component efficiencies, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.2.12.4.(1)

Objective
OE1

Attributions
[F98-OE1.1]

Intent(s)

Intent 1. To limit the probability that service water heating equipment used to provide comfort conditioning will not meet minimum acceptable levels of performance, which could lead to inefficient equipment, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 5.4.1.1.(1)

Intent(s)

Intent 1. To direct Code users to Part 8.

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Provision: 5.4.1.2.(1)

Objective
OE1

Attributions
[F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that heating, ventilating and air-conditioning equipment and systems designed using the performance compliance path will not meet minimum regulated levels of efficiency, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for heating, ventilation and air-conditioning, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To limit the application of the performance compliance path for heating, ventilating and air-conditioning systems.

Provision: 5.4.1.2.(2)

Intent(s)
Intent 1. To limit the application of Sentence 5.4.1.1.(1) and require that back-up systems meet the prescriptive requirements of the Code, and not a lesser standard, on the basis that they are required for emergencies.

Provision: 6.1.1.1.(1)

Intent(s)
Intent 1. To state the scope of Part 6.

Provision: 6.1.1.2.(1)

Intent(s)
Intent 1. To state the application of Part 6.

Provision: 6.1.1.3.(1)

Intent(s)
Intent 1. To state the methods of compliance with Part 6.

Intent 2. To direct Code users to Section 6.2. for the prescriptive path requirements and Section 6.4. for the building energy performance compliance path requirements.

Provision: 6.1.1.3.(2)

Intent(s)

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Intent 1. To limit the method of compliance for back-up systems to the prescriptive path, on the basis that they provide minimum service water heating levels in the event the primary system is compromised.

Provision: 6.1.1.4.(1)

Intent(s)

Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 6.2.1.1.(1)

Objective

OE1

Attributions

[F96, F98-OE1.1]

Intent(s)

Intent 1. To limit the probability that service water systems will not meet minimum regulated levels of efficiency, which could lead to inefficient equipment, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.2.1.(1)

Objective

OE1

Attributions

[F96, F98-OE1.1]

Intent(s)

Intent 1. To limit the probability that service water heating equipment and pool heaters will not meet minimum acceptable levels of performance, which could lead to inefficient equipment, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.2.2.(1)

Objective

OE1

Attributions

[F93, F96-OE1.1]

Intent(s)

Intent 1. To limit the probability that hot water storage tanks will be insufficiently insulated, which could lead to uncontrolled thermal transfer from the storage tank, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

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Intent 1. To exempt hot water storage tanks that meet the insulation levels prescribed by acceptable standards and acts from the insulation requirements stated in Sentence 6.2.2.2.(1).

Provision: 6.2.2.2.(2)

Objective
OE1

Attributions
[F93, F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that storage tank insulation will be damaged, which could lead to uncontrolled thermal transfer from the storage tank, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.2.3.(1)

Objective
OE1

Attributions
[F96, F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that solar service water heating equipment will not be appropriately installed or designed, which could lead to an inability to meet minimum acceptable levels of performance, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.2.4.(1)

Objective
OE1

Attributions
[F95, F96, F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that

- designs with a large demand for combined service water heating and space-heating will use a combination system, which could lead to inefficient equipment and systems, or
- combination systems will be used where the space-heating load exceeds the service water heating load.

This is to limit the probability of unnecessary consumption of energy for service water heating and/or space heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 6.2.2.4.(2)

Objective
OE1

Attributions
[F95, F96, F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that combination equipment used for service water heating and space heating will not meet minimum acceptable levels of performance, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for service water heating and/or space heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.2.5.(1)

Objective
OE1

Attributions
[F95, F96, F98, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that space-heating equipment used to provide indirect service water heating solely or in combination with space heating will not meet the minimum acceptable levels of performance

- for service water heating equipment, which could lead to inefficient service water heating equipment and/or systems, which could lead to unnecessary consumption of energy for service water heating, and
- for space-heating equipment, which could lead to inefficient space-heating equipment and/or systems, which could lead to unnecessary consumption of energy for space heating.

This is to limit the probability of excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.3.1.(1)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)
Intent 1. To limit the probability that piping in service water heating systems will be insufficiently insulated, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 6.2.3.1.(2)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that there will be insufficient insulation for piping in service water heating systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.3.1.(3)

Intent(s)

Intent 1. To supersede the insulation requirement of Sentence 6.2.3.1.(1) and allow a decrease in the thickness of insulation where its conductivity is lower than that prescribed in Table 6.2.3.1.

Provision: 6.2.3.1.(4)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermal conductivity of pipe insulation at the mean temperature rating will not be determined in accordance with an acceptable standard, which could lead to insufficient insulation for piping in service water heating systems, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.3.1.(5)

Objective
OE1

Attributions
[F92, F93-OE1.1]

Intent(s)

Intent 1. To limit the probability that piping in service water heating systems will be insufficiently insulated, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 6.2.3.1.(6)

Objective
OE1.1

Attributions
[F93, F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulation on hot service water piping will be undersized, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.3.1.(7)

Objective
OE1.1

Attributions
[F93, F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that insulation on hot service water piping located where it may be damaged or subject to weathering will be unprotected, which could lead to insulation being damaged or weathered, which could lead to reduced performance of the insulation, which could lead to uncontrolled thermal losses or gains, which could lead to unnecessary consumption of energy for heating and cooling, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.3.1.(8)

Attributions
[F93, F95, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the thermal performance of the installed insulation will be lower than that specified by the manufacturer, which could lead to uncontrolled thermal transfer losses or gains, which could lead to unnecessary demand and/or consumption of energy for heating and cooling, which could lead to inefficient heating and cooling systems, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.4.1.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that service water heating systems with storage tanks will not be equipped with automatic temperature controls within a range required for intended use, which could lead to

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overheating of the water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.4.2.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that service water heating systems and pipe heating elements will not be equipped with shut-offs, which could lead to unwarranted operation of the system, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To exempt systems with a small storage capacity from the requirements for shut-offs, on the basis that the operation of these systems will not significantly impact the consumption of energy.

Provision: 6.2.4.3.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that heating elements to maintain temperature along service water pipes will not be equipped with automatic controls to maintain the temperature within a range required for intended use, which could lead to overheating of the water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.5.1.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that service water heating systems with less than 50% of the total design flow that discharges water at a temperature higher than 60°C will heat greater than that portion of the total design flow, which could lead to overheating of the remainder of the design flow to greater than the service temperature required, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 6.2.6.1.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that individual shower heads will not be equipped with an integral means of limiting the water flow, which could lead to excessive use of hot service water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt from the water flow limits, emergency eye washes and emergency showers that are required to have higher flow rates to fulfill their intended use, which relates to safety.

Provision: 6.2.6.1.(2)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that individual shower heads will discharge hot service water when not in use, which could lead to excessive use of hot service water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.6.2.(1)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that lavatories will not be equipped with an integral means of limiting the hot water flow, which could lead to excessive use of hot service water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt lavatories in health care facilities and emergency eye washes from the requirements of having an integral means of limiting water flow, on the basis that these facilities are required to have higher flow rates to fulfill their intended use, which relates to safety.

Provision: 6.2.6.2.(2)

Objective
OE1

Attributions
[F96-OE1.1]

Intent(s)
Intent 1. To limit the probability that lavatories in assembly occupancies will discharge hot service water when not in use, which could lead to excessive use of hot service water, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.7.1.(1)

Objective
OE1

Attributions
[F95, F96, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that pool heaters will not be equipped with an adequate method to be easily shut off and restarted, which could lead to operation of the heater when not warranted, which could lead to unnecessary consumption of energy for pool water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.7.1.(2)

Objective
OE1

Attributions
[F95, F96, F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that pool heaters and pumps will not be equipped with automatic controls, which could lead to their operation when not in use, which could lead to unnecessary consumption of energy for pool operation and water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To exempt from the requirement for automatic controls, pool heaters and pumps that are required to operate continuously for public health reasons.

Provision: 6.2.7.2.(1)

Objective
OE1

Attributions
[F95-OE1.1]

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Intent(s)

Intent 1. To limit the probability that heated outdoor pools and tubs will not be equipped with covers, which could lead to excessive heat loss, which could lead to unnecessary operation of heaters, which could lead to unnecessary consumption of energy for water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.7.2.(2)

Objective
OE1

Attributions
[F95-OE1.1]

Intent(s)

Intent 1. To limit the probability that covers for pools and tubs heated to high temperatures will have excessive thermal transmittance, which could lead to excessive heat loss, which could lead to unnecessary operation of heaters, which could lead to unnecessary consumption of energy for water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.8.1.(1)

Objective
OE1

Attributions
[F97, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that constant-speed pressure booster systems will have insufficient water storage volume, which could lead to unnecessary operation or short-cycling of the booster pumps, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.8.1.(2)

Objective
OE1

Attributions
[F97, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that variable-speed pressure booster systems will have insufficient water storage volume, which could lead to unnecessary operation or short-cycling of the booster pumps, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 6.2.8.2.(1)

Objective
OE1

Attributions
[F96, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that operation of pressure booster systems will not be controlled by at least one pressure sensor, which could lead to unnecessary operation of the booster pumps or pressure boosting of the supplied service water, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.2.8.2.(2)

Objective
OE1

Attributions
[F96, F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that pressure-reducing devices will be inappropriately located, which could lead to unnecessary pressure reduction of all service water supplied, which could lead to unnecessary consumption of energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 6.4.1.1.(1)

Intent(s)

Intent 1. To direct Code users to Part 8.

Provision: 6.4.1.2.(1)

Objective
OE1

Attributions
[F98, F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that service water heating equipment and systems designed using the performance compliance path will not meet minimum regulated levels of efficiency, which could lead to inefficient equipment and/or systems, which could lead to unnecessary consumption of energy for service water heating, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the application of the performance compliance path for service water heating systems.

Provision: 6.4.1.2.(2)

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Intent(s)

Intent 1. To limit the application of Sentence 6.4.1.1.(1) and require that back-up systems meet the prescriptive requirements of the Code, and not a lesser standard, on the basis that they are required for emergencies.

Provision: 7.1.1.1.(1)

Intent(s)

Intent 1. To state the scope of Part 7.

Provision: 7.1.1.2.(1)

Intent(s)

Intent 1. To state the application of Part 7.

Provision: 7.1.1.3.(1)

Intent(s)

Intent 1. To state the methods of compliance with Part 7.

Intent 2. To direct Code users to Section 7.2. for the prescriptive path requirements and Section 7.4. for the building energy performance compliance path requirements.

Provision: 7.1.1.4.(1)

Intent(s)

Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 7.2.1.1.(1)

Objective

OE1

Attributions

[F97-OE1.1]

Intent(s)

Intent 1. To limit the probability that a means to monitor the electrical energy consumption of certain energy end uses will not be installed, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.2.1.1.(2)

Objective

OE1

Attributions

[F97-OE1.1]

Intent(s)

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Intent 1. To limit the probability that a means to separately monitor the electrical energy consumption of the total building and of each individual dwelling unit will not be installed, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.2.2.1.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that line voltage drops in feeder conductors will exceed acceptable levels, which could lead to unnecessary energy losses in the electrical distribution system, which could lead to an inefficient electrical distribution system, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.2.2.2.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that line voltage drops in branch circuit conductors will exceed acceptable levels, which could lead to unnecessary energy losses in the electrical distribution system, which could lead to an inefficient electrical distribution system, could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.2.3.1.(1)

Objective
OE1

Attributions
[F97, F98-OE1.1]

Intent(s)
Intent 1. To limit the probability of high losses from transformers, which could lead to the use of inefficient equipment, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.2.4.1.(1)

Objective
OE1

Attributions
[F97, F98, F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that motors will not meet minimum acceptable levels of performance, which could lead to inefficient equipment and/or systems, which could lead to unnecessary energy demand and/or consumption of energy of electrical equipment and devices, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 7.4.1.1.(1)

Intent(s)

Intent 1. To direct Code users to Part 8.

Provision: 8.1.1.1.(1)

Intent(s)

Intent 1. To state the scope of Part 8.

Provision: 8.1.1.2.(1)

Intent(s)

Intent 1. To state the application of Part 8.

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the Part 8 compliance path will be used when there is insufficient information to accurately model the reference and proposed buildings, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.1.1.2.(2)

Intent(s)

Intent 1. To limit the application of Sentence 8.1.1.2.(1) and direct Code users to the applicable prescriptive requirements in Sections 3.2., 4.2., 5.2., 6.2. and 7.2. when there is insufficient information about certain components, materials or assemblies.

Provision: 8.1.1.2.(3)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that buildings whose construction varies from the proposed design will not have their performance reassessed, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.1.1.2.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the Part 8 compliance path will be used for a group of stand-alone structures, which could lead to underestimation of the energy used by each building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.1.1.2.(5)

Intent(s)
Intent 1. To expand the application of Sentence 8.1.1.2.(4) and allow the Part 8 compliance path to be used for single structures divided by firewalls into buildings, on the basis that they can be accurately modeled for energy performance.

Provision: 8.1.1.3.(1)

Intent(s)
Intent 1. To direct Code users to Article 1.4.1.2. of Division A for the definitions of italicized words.

Provision: 8.4.1.1.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the building energy target will not correspond to the energy use of the building when designed to the acceptable solution set by the prescriptive path, which could lead to overestimation of the building energy target, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.1.(2)

Objective
OE1

Attributions
[F92, F93, F94, F95, F96, F97, F98, F99, F100-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the annual energy consumption for the proposed building will give credit for construction techniques or building components that are more energy-efficient than the levels set by the prescriptive path but whose performance level is dependant on occupant behaviour (which cannot be mandated by the Code), or cannot be quantified, which could lead to underestimation of the annual energy consumption, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.1.(3)

Intent(s)

Intent 1. To limit the application of Sentence 8.4.1.1.(1) and allow the modeling calculations to be simplified for exterior lighting that meets or exceeds the prescriptive levels, on the basis that indoor loads due to exterior lighting equipment have no impact on the overall energy use of the rest of the building.

Provision: 8.4.1.2.(1)

Intent(s)

Intent 1. To state the basis of compliance with Part 8.

Provision: 8.4.1.2.(2)

Objective

OE1

Attributions

[F92, F93, F94, F95, F96, F97, F98, F99, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that the energy consumption of the proposed building will exceed the energy consumption of the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.2.(3)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating system of the reference or proposed building will not be capable of meeting the heating demands of the building, which could lead to underestimation of the energy used for heating, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.1.2.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the cooling system of the reference or proposed building will not meet the cooling demands of the building to the same level, which could lead to underestimation of the energy used for cooling, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.2.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating or cooling systems of the reference or proposed building will not be capable of meeting the heating or cooling demands of the building, which could lead to underestimation of the energy used for heating or cooling, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.3.(1)

Intent(s)

Intent 1. To direct Code users to Sections 3.4., 4.4., 5.4., 6.4. and 7.4. for limitations on the compliance calculations.

Provision: 8.4.1.4.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that, where additions are involved, the building energy target of the reference building or the annual energy consumption of the proposed building will be incorrectly calculated, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.1.4.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that, where additions are involved, the characteristics of existing components will not be accurate, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.1.4.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that, where additions are involved, the existing primary and secondary systems will be incorrectly modeled, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.1.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that assessment of conformance will be performed incorrectly, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.2.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the annual energy consumption of the proposed building will not take into account the impact of all components that are addressed by the Code, which could lead to

underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.2.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the calculations will be performed for a period that is insufficiently long to represent the annual performance or over intervals that are too large to provide enough accuracy in the result, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.2.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the schedules and climatic data for the energy model will use intervals that are too large to provide enough accuracy in the result, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.2.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the calculation methods will not be performed to a level set by an industry-recognized standard, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.2.(5)

Intent(s)
Intent 1. To exempt redundant or back-up equipment whose operation does not impact the building energy consumption from the modeling requirements of Sentence 8.4.2.2.(1).

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Provision: 8.4.2.3.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that inappropriate or insufficient climatic data will be used, which could lead to inaccurate modeling, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.3.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that inappropriate, insufficient or incomplete climatic data will be used, which could lead to inaccurate modeling, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.4.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effects of thermal mass will not be accounted for in the energy model, which could lead to inaccurate modeling, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.5.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the model will not perform a dynamic calculation of space temperatures that accounts for all effects, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by

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the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.6.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the model will not account for adjacent thermal blocks with significant temperature differences, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.6.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that heat transfer calculation between thermal blocks not fully separated by solid partitions or walls will use an inappropriate value of equivalent heat transfer coefficient, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.7.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the model will not account for all internal and service water heating loads that affect the energy performance, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.2.7.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the sensible and latent loads will not be accounted for in the energy model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.7.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the interior loads will not be properly adjusted for each time interval based on the appropriate schedule, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.7.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the calculation of sensible loads due to lights will not take into account all effects, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.7.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that other equipment located within a conditioned space that affect the energy performance of the building will not be accounted for in the energy model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the energy model will not account for heat transfer through all relevant building envelope elements for each thermal block, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent 2. To limit the probability that the dynamic response due to thermal characteristics of the building envelope assemblies will not be accounted for in the energy model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effect of solar radiation on heat transfer through walls and roofs will not be accounted for in the energy model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the heat transfer through fenestration, including skylights, will not be appropriately accounted for in the energy model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the

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energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the impacts of solar radiation through fenestration on the heating and cooling loads will not be appropriately accounted for, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the insulated surface area of roof assemblies will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the insulated surface area of exterior above-ground wall assemblies will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.2.8.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulated surface area of above-ground exterior floor assemblies will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulated surface areas of roof assemblies in contact with the ground will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(9)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the insulated surface areas of wall assemblies in contact with the ground will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.8.(10)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the insulated surface area of floor assemblies in contact with the ground will be incorrectly determined, which could lead to incorrect modeling of the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive consumption of energy by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.9.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the model will not account for air leakage through the building envelope, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.9.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the adjusted air leakage rate of the building envelope will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.10.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the HVAC system of the reference building will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.2.10.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that energy model will not take into account the effects of terminal devices, secondary and primary systems, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.10.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the compliance calculations for secondary systems will not take into account all relevant factors, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.10.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the energy model will not take into account how many hours the primary system, secondary system and terminal device loads are not met, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.2.10.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the energy model will not take into account the efficiency and capacity of HVAC equipment as a function of part load and all relevant parameters, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.1.(1)

Intent(s)

Intent 1. To state the application of Subsection 8.4.3.

Provision: 8.4.3.1.(2)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the energy model will not accurately reflect the specifications of the proposed building, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.1.(3)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that all energy uses addressed in the prescriptive path will not be accounted for in the model, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.1.(4)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the energy model of the proposed building will not accurately reflect the type of thermal conditioning installed, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.3.1.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that building systems or parts thereof that have not been fully specified will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.1.(6)

Intent(s)

Intent 1. To exempt from the application of Sentence 8.4.3.1.(2), envelope assemblies that, due to their small relative size and proximity to similar assemblies, can be alternatively modeled without any significant impact on the energy result.

Provision: 8.4.3.1.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that manually operated fenestration shading devices, whose operation at the appropriate times is unreliable as it requires occupant interaction, will be modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.1.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that permanent shading devices will not be included in the model, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.3.2.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the operating schedules will not be representative of the intended use for the building or space, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.2.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the internal and service water heating loads and illuminance levels will not be representative of the intended use for the building or space, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.2.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To supersede the requirement of Sentence 8.4.4.1.(4) and permit a different set-point temperature in the reference and proposed buildings, where the set-point temperature is lower than 18°C.

Provision: 8.4.3.3.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the solar absorptance of an envelope component will be set at an inappropriate value, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.3.3.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the calculation of shading will result in an inappropriate value, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.3.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the air leakage of building envelope components will be set at an inappropriate value, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)
Intent 1. To direct Code users to Sentence 8.4.2.9.(2) for the calculation of the adjustment of the normalized air leakage rate.

Provision: 8.4.3.3.(4)

Intent(s)
Intent 1. To supersede the requirements of Sentence 8.4.3.3.(3) and allow the use of the normalized air leakage rate at a pressure differential of 75 Pa and the flow exponent from the test when the air leakage rate is determined in accordance with Article 3.2.4.2.

Provision: 8.4.3.4.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the installed lighting power density of dwelling units will be set at too low or too high a value, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.3.4.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the occupancy-sensing mechanism for installed interior lighting power will be incorrectly set, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.4.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the daylight calculations will be incorrectly performed, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.4.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the illumination set-point of photocontrols will be improperly set in the proposed building, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.5.(1)

Intent(s)
Intent 1. To state the application of Sentences 8.4.3.5.(2) to 8.4.3.5.(5).

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Provision: 8.4.3.5.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that purchased energy used for heating will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.5.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that purchased energy used for cooling will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.5.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that purchased energy used for service water heating will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.5.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the operating schedule, priority of use and other operational characteristics of the purchased energy will not be included in the model, which could lead to inaccuracy of

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the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.6.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the peak outdoor air ventilation rates will be set at too low or too high a value, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.7.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the throttling range of space-conditioning equipment will be inappropriately set, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.8.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load performance curves will be inappropriately set, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.8.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the part-load performance curves will be inappropriately set, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.3.9.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that ice-making loads will be inappropriately set, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.1.(1)

Intent(s)
Intent 1. To state the application of Subsection 8.4.4.

Provision: 8.4.4.1.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the components and systems of the reference building will not be modeled in accordance with the prescriptive requirements, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.1.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that all energy uses addressed in the prescriptive path will not be accounted for in the model, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.1.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the characteristics of the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.1.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the presence or absence of heating and/or cooling systems in each conditioned thermal block of the reference building will differ from what was specified for the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.1.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the climatic data used in the compliance calculations for the reference building will differ from those used for the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.1.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effect of part-load operation on equipment performance will not be accounted for in the compliance calculations of the reference building, which could lead to inaccuracy

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of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.2.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the operating schedules for the reference building will differ from those for the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.2.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the internal and service water heating loads of the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.2.(3)

Intent(s)
Intent 1. To supersede the requirement of Sentence 8.4.4.1.(4) and permit a different set-point temperature in the reference and proposed buildings, where the set-point temperature is lower than 18°C, and the heating system to be installed in the proposed building is not capable of providing an appreciably higher set-point temperature than that shown in the specifications.

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the reference building's thermal characteristics will be based on the prescriptive semi-heated requirements that are derived based on heating degree-days at 15°C of the building's location, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.3.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the solar absorptance of each opaque envelope assembly of the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.3.(2)

Objective
OE1

Attributions
8.4.4.3.(2)(a) **[F99-OE1.1]**

Intent(s)

Intent 1. To limit the probability that the solar absorptance of roof assemblies of the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Intent(s)

Intent 1. To supersede the requirement of Sentence 8.4.4.3.(1) and allow the solar absorptance of roof assemblies to be modeled in the reference building in order to give a credit to proposed buildings that have strategies for the solar absorptance of roof assemblies.

Provision: 8.4.4.3.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fenestration and door areas of the reference building will not be distributed consistently with their distribution in the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.3.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that permanent shading devices and projections will be included in the energy model of the reference building, which could lead to reduced heat gains, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.3.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that exterior shading provided by a nearby structure or building will be modeled differently in the reference and proposed buildings, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.3.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the air leakage rate of the reference building will differ from the default value used for the proposed building as determined in Sentence 8.4.3.3.(3), which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.3.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that heat transfer through interior partitions in the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.3.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the thermal and optical properties of the fenestration of the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.4.(1)

Intent(s)
Intent 1. To require that the reference building be modeled with light-weight construction so that the effects of thermal mass in the proposed building can be taken into account in accordance with Article 8.4.2.4.

Provision: 8.4.4.4.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the thermal characteristics of the space of the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the interior lighting power density of the reference building will not be determined in accordance with the prescriptive requirements, which could lead to overestimation of the installed interior lighting power in the reference building, which could lead to inaccuracy of the

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model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the installed lighting power density of dwelling units will be set at too high a value, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the occupancy control or personal control factors for installed interior lighting power will be inappropriately selected, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the proportions of radiant and convective heat and the percentages of heat from lights going directly to return air in the reference building will differ from those in the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(5)

Objective
OE1

Attributions
[F99-OE1.1]

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Intent(s)

Intent 1. To limit the probability that the total fenestration area of each thermal block will be inadequately defined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that sidelighted areas in the reference building will be inadequately defined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the daylighted areas under skylights in the reference building will be inadequately defined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the combined input power within the daylighted areas will be incorrectly calculated in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.5.(9)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the energy model will not properly account for the presence of photocontrols in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(10)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the interior illumination levels due to daylighting in each thermal block will be incorrectly calculated in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(11)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the illumination set-point of photocontrols will be improperly set in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.5.(12)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the effect of daylighting will be improperly accounted for in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of

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the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.6.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the heating systems of the reference building will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.6.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling systems of the reference building will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.6.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the service water heating systems of the reference building will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.6.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the operating schedule, priority of use and other operational characteristics of the purchased energy of the proposed building will not be correctly captured by the energy model of the reference building, which could lead to inaccuracy of the model, which could lead to underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.7.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate type of HVAC system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.7.(2)

Intent(s)
Intent 1. To exempt from the requirement to assign an HVAC system stated in Sentence 8.4.4.7.(1), areas that are small in size relative to the total building and whose loads are accounted for in an adjacent space with similar characteristics.

Provision: 8.4.4.7.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate type of HVAC system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.7.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate type of HVAC system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the

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energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.8.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the heating equipment of the reference building will be excessively oversized, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.8.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling equipment of the reference building will be excessively oversized, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate type of heating system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(2)

Intent(s)
Intent 1. To direct Code users to Article 8.4.4.6. for the requirements on purchased energy.

Provision: 8.4.4.9.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating capacity of each type of equipment will be inappropriately set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the energy type of the heating system of the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that

- the heating capacities of the heating equipment of the reference building will not match the ratio of the heating equipment capacity allocation of the heating equipment of the proposed building, and
- the operating schedule, priority of use, and other operational characteristics will not match those for the energy types used in the proposed building.

This is to limit the probability of inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.9.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating plant for hydronic systems will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that furnaces will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.9.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that heating equipment performance characteristics as a function of part-load will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that an inappropriate type of cooling system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the

energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(2)

Intent(s)

Intent 1. To direct Code users to Article 8.4.4.6. for the requirements on purchased energy.

Provision: 8.4.4.10.(3)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the energy type of the cooling system of the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(4)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that

- the cooling capacities of the cooling equipment of the reference building will not match the ratio of the cooling equipment capacity allocation of the cooling equipment of the proposed building, and
- the operating schedule, priority of use, and other operational characteristics will not match those for the energy types used in the proposed building.

This is to limit the probability of inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(5)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that cooling equipment performance characteristics as a function of part-load will be incorrectly modeled for the reference building, which could lead to inaccuracy of the

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model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling plant for hydronic systems will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling capacity of each type of equipment will be inappropriately set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling capacity of direct-expansion systems in the reference building will be incorrectly determined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.10.(9)

Objective
OE1

Attributions
[F99, F100-OE1.1]

Intent(s)

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Intent 1. To limit the probability that heat rejection to the atmosphere from water-cooled systems will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building and underestimation of the rejection of reusable waste energy, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the axial-fan, direct-contact cooling tower for the reference building will have inappropriate characteristices, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that small-capacity cooling towers will be modeled with more than one cell in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that larger-capacity cooling towers will be modeled with an inappropriate number of cells in the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.11.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the pumping system will not be modeled at a constant speed of operation, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the pumping flow rate will be set at an inadequate value, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fan of each cooling tower cell will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.11.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling tower equipment performance characteristics will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation

of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.12.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the reference building's HVAC systems will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.13.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate HVAC system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.13.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that an inappropriate HVAC system will be selected for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.14.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the total static head and efficiency of the hydronic pump of the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.14.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the shaft power demand of the hydronic pump in the reference building will not be equivalent to the combined shaft power demand of the pump in the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.14.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the total static head and efficiency of the hydronic pump of the reference building will not be set at reasonable values, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.14.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the pump-power-versus-flow-rate characteristics will be inappropriately set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.14.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that variable-flow pumps will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.14.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the capacity and performance of the heating and cooling equipment of the secondary system for equipment whose rated performance includes pumps is not adjusted to meet the minimum efficiency levels prescribed by Tables 5.2.12.1.-A to 5.2.12.1.-P, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.15.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the amount of outdoor air supplied to the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.15.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that demand control ventilation strategies will be incorrectly modeled for the reference building, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.16.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the temperature schedules for the reference building will not account for the effects of in-floor, in-ceiling or in-wall radiant heating and/or cooling, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.16.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the thermostat throttling ranges of heating and cooling equipment in the reference building will differ from those of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.17.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the space exhaust fans of the proposed building will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.17.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fan-power-versus-flow-rate characteristics will be inappropriately set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.17.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that individual fans will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.17.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that individual variable-air-volume fans will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.17.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that individual variable-air-volume fans will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy

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used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the supply airflow rate of secondary systems will be incorrectly determined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the supply airflow rate of each of the ventilated thermal blocks will be incorrectly determined, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the supply air temperature, supply fan and return fan for Systems 1 through 5 and 7 of Table 8.4.4.7.-B will be incorrectly set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the supply air temperature, supply fan and return fan for System 6 of Table 8.4.4.7.-B will be incorrectly set, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fan power demand of the reference building will differ from that of the proposed building for fans with low power demand, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.18.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the capacity and performance of the heating and cooling equipment of the secondary system for equipment whose performance includes fan systems is not adjusted to meet the minimum efficiency levels prescribed by Tables 5.2.12.1.-A to 5.2.12.1.-P, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.19.(1)

Objective
OE1

Attributions
[F99, F100-OE1.1]

Intent(s)

Intent 1. To limit the probability that the HVAC systems of the reference building will not have energy recovery systems complying with the prescriptive requirements, which could lead to unnecessary rejection of reusable waste energy, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.19.(2)

Objective
OE1

Attributions
[F99, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that energy recovered from an HVAC system's exhaust air will not be used to preheat outside air, which could lead to unnecessary rejection of reusable waste energy, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.19.(3)

Objective
OE1

Attributions
[F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that water-cooled water chillers serving ice rinks and curling rinks will not be equipped with heat-recovery systems that will be modeled appropriately, which could lead to unnecessary rejection of reusable waste energy, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the service water heating system of the reference building will differ from that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the energy source of the service water heating system of the reference building will not be compatible with that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the energy source of the service water heating system of the reference building will not be compatible with that of the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that

- the heating capacities of the service water heating equipment of the reference building will not match the ratio of the service water heating equipment capacity allocation of the proposed building, and
- the operating schedule, priority of use, and other operational characteristics will not match those for the energy types used in the proposed building.

This is to limit the probability of inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load characteristics of service water heating equipment will be incorrectly modeled for the reference building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.4.20.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the supply temperature of the service hot water will be set at too high a level, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the storage temperature of the service hot water tank will be set at too high a level, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the number of water heaters in the reference building will not match that in the proposed building, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building or underestimation of the energy used by the proposed building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.4.20.(9)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the circulation pumps of the reference building will be incorrectly modeled, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.5.1.(1)

Intent(s)

Intent 1. To direct Code users to Articles 8.4.5.2. to 8.4.5.9. for the calculation of part-load performance curves for the reference building's systems.

Provision: 8.4.5.2.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fuel consumption at part-load conditions of the reference condensing or non-condensing boiler will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.2.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fuel heating part-load efficiency curve of the reference condensing or non-condensing boiler will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.2.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the fuel heating part-load efficiency curve of the reference modulating boiler will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.5.3.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel consumption at part-load conditions of the reference condensing or atmospheric furnace will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.3.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel heating part-load efficiency curve of the reference condensing or atmospheric furnace will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.3.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel heating part-load efficiency curve of the reference modulating furnace will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(1)

Intent(s)
Intent 1. To direct Code users to Sentence 8.4.5.4.(2) for the adjustment for cooling capacity and to Sentence 8.4.5.4.(4) for the adjustment for power draw to be applied to the part-load performance characteristics of the reference electric direct-expansion coil cooling equipment.

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Provision: 8.4.5.4.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the available total cooling capacity of the reference electric direct-expansion coil will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling capacity adjustment of the reference electric direct-expansion coil will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the power draw at specified operating conditions of the reference electric direct-expansion coil will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency due to changes in coil load will be incorrectly calculated, which could lead to inaccuracy of the model,

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which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load ratio based on available capacity of the reference electric direct-expansion coil will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency due to environmental variables will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.4.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the inappropriate outdoor-air dry-bulb temperature will be used in cases where an air-cooled unit uses an evaporative condenser, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(1)

Intent(s)
Intent 1. To direct Code users to Sentence 8.4.5.5.(2) for the adjustment for cooling capacity and to Sentence 8.4.5.5.(4) for the adjustment for power draw to be applied to the part-load performance characteristics of the reference electric chiller.

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Provision: 8.4.5.5.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the available total cooling capacity of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling capacity adjustment of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the power draw at specified operating conditions of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency, due to changes in load, of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference

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building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load ratio based on available capacity of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.5.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency, due to environmental variables, of the reference electric chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.6.(1)

Intent(s)
Intent 1. To direct Code users to Sentence 8.4.5.6.(2) for the adjustment for cooling capacity to be applied to the part-load performance characteristics of the reference cooling tower.

Provision: 8.4.5.6.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the available total cooling capacity of the reference cooling tower will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.5.6.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the ratio of available capacity to rated capacity of the reference cooling tower will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.6.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the intermediate capacity curve based on range and approach of the reference cooling tower will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.6.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the range of the reference cooling tower will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.6.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the approach of the reference cooling tower will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy

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used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.7.(1)

Intent(s)

Intent 1. To direct Code users to Sentence 8.4.5.7.(2) for the adjustment for heating capacity and to Sentence 8.4.5.7.(4) for the adjustment for power draw to be applied to the heating part-load performance characteristics of the reference electric air-source heat pump.

Provision: 8.4.5.7.(2)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the available heating capacity of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.7.(3)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the heating capacity adjustment of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.7.(4)

Objective

OE1

Attributions

[F99-OE1.1]

Intent(s)

Intent 1. To limit the probability that the power draw of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.5.7.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency, due to changes in heat pump load, of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.7.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load ratio based on available capacity of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.7.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the electric input ratio adjustment to the rated efficiency, due to environmental variables, of the reference electric air-source heat pump will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(1)

Intent(s)
Intent 1. To direct Code users to Sentence 8.4.5.8.(2) for the adjustment for cooling capacity and to Sentences 8.4.5.8.(4) and 8.4.5.8.(8) for the adjustment for fuel consumption to be applied to the part-load performance characteristics of the reference absorption chiller.

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Provision: 8.4.5.8.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the available total cooling capacity of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the cooling capacity adjustment of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel consumption at specified operating conditions of a reference steam-driven, single- or double-effect absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(5)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel input ratio adjustment to the rated efficiency, due to changes in load, of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy

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of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(6)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the part-load ratio based on available capacity of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(7)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel input ratio adjustment to the rated efficiency, due to environmental variables, of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(8)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel consumption at specified operating conditions of a reference direct-fired, double-effect absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 8.4.5.8.(9)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel input ratio adjustment to the rated efficiency, due to changes in load, of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(10)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the first fuel input ratio adjustment to the rated efficiency, due to environmental variables, of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.8.(11)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the second fuel input ratio adjustment to the rated efficiency, due to environmental variables, of the reference absorption chiller will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.9.(1)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)

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Intent 1. To limit the probability that the fuel consumption at part-load conditions of the reference fuel-fired service water heater will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 8.4.5.9.(2)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the fuel heating part-load efficiency curve of the reference fuel-fired service water heater will be incorrectly calculated, which could lead to inaccuracy of the model, which could lead to overestimation of the energy used by the reference building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 10.1.1.1.(1)

Intent(s)
Intent 1. To state the scope of Part 10.

Provision: 10.1.1.2.(1)

Intent(s)
Intent 1. To state the application of Part 10.

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the Part 10 compliance path will be used when there is insufficient information to accurately model the reference and proposed buildings, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 10.1.1.2.(2)

Intent(s)
Intent 1. To limit the application of Sentence 10.1.1.2.(1) and direct Code users to the applicable prescriptive requirements in Sections 3.2., 4.2., 5.2., 6.2. and 7.2. when there is insufficient information about certain building components, materials or assemblies.

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Provision: 10.1.1.2.(3)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that buildings whose construction deviates from the proposed design will not have their performance reassessed, which could lead to underestimation of the energy used by the building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 10.1.1.2.(4)

Objective
OE1

Attributions
[F99-OE1.1]

Intent(s)
Intent 1. To limit the probability that the Part 10 compliance path will be used for a group of stand-alone structures, which could lead to underestimation of the energy used by each building, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

Provision: 10.1.1.2.(5)

Intent(s)
Intent 1. To expand the application of Sentence 10.1.1.2.(4) and allow the Part 10 compliance path to be used for single structures divided by firewalls into multiple buildings, on the basis that they can be accurately modeled for energy performance.

Provision: 10.1.2.1.(1)

Intent(s)
Intent 1. To state the basis of compliance with Part 10.

Objective
OE1

Attributions
[F92, F93, F94, F95, F96, F97, F98, F99, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that the percent building energy target for each tier will exceed the level set in Table 10.1.2.1., or the percent improvement for each tier will be less than the level set in Table 10.1.2.1., which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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Provision: 10.1.2.1.(2)

Objective
OE1

Attributions
[F92, F93, F94, F95, F96, F97, F98, F99, F100-OE1.1]

Intent(s)
Intent 1. To limit the probability that the percent building energy target or the percent improvement will be incorrectly calculated, which could lead to excessive use of energy, which could lead to an unacceptable effect on the environment.

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